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Aging behavior of LiFePO₄-based battery cells at stack level: A Second-Life cycling studyDaniel Aeppli^{*}, Erwin Hack, Marcel Held

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ABSTRACT

This study investigates capacity degradation in 100 Ah lithium iron phosphate (LiFePO₄, LFP) battery cells during First-Life and Second-Life use. In a laboratory-based case study, cells achieved up to 9600 cycles, with a maximum loss of 42 Ah over 7 years of Second-Life use and a total loss of 56 Ah over >14 years of total cell age. These results highlight the long-term viability of LFP cells for battery energy storage systems. A power law model based on the first 10 data points effectively described capacity trajectories and identified early signs of accelerated degradation. One cell deviated from the predicted progression at cycle #8000, corresponding to a State-of-Health (SoH) of 48 %. Crossing the ± 3 standard deviation threshold in the capacity residuals marked the End-of-Second-Life criterion. Incremental Capacity Analysis indicated additional aging mechanisms, such as Loss of Active Material and increased faradaic polarization. Cells with lower initial capacity degraded faster at 70 % SoH, suggesting a History effect overlapping with temperature-dependent aging. Only cells with high initial capacity at 70 % SoH exhibited a linear Arrhenius trend, while cells with low initial capacity degraded slightly faster at 10 °C than at 30 °C, suggesting different aging mechanisms. Within the stack, temperatures recorded in a 10 °C climate chamber reached 22 °C, challenging the reliability of setpoints. Spatial temperature variations suggested non-uniform degradation patterns. These findings underscore the importance of considering capacity history, thermal gradients, and early diagnostic indicators when assessing Second-Life battery performance.

1. Introduction

1.1. General introduction

The global demand for Lithium-ion batteries is driven by consumer electronics, electric vehicles (EVs) and battery energy storage systems (BESS) [1]. As lithium and transition metal resources are limited, circular economy practices have become important [2,3]. A key focus in the energy sector is reusing End-of-Life (EOL) traction batteries in Second-Life (SL) applications, such as BESS, while ensuring safety and durability [4,5]. The economic feasibility of SL BESS is typically higher for systems larger than 1 MWh, with implementation costs reduced through accurate capacity fade predictions [6].

Longevity is crucial for BESS, with profitability relying on a system lifetime of up to 20 years or around 6000 Full Equivalent Cycles (FECs) for home energy storage [7]. FEC is defined as the charge-throughput (Ah) divided by twice the nominal capacity (Ah).

Cell matching is crucial for battery pack longevity. Although manufacturers group cells into submodules to minimize variations in

capacity and internal resistance, a tailored Battery Management System (BMS) remains essential to prevent imbalance issues, a primary cause of degradation [8–10].

Cycle life is defined as the number of full discharge-charge cycles until a cell reaches its EOL, usually when its capacity drops to 80 % of the nominal capacity value. Rimpas et al. [11] suggested limiting a battery's First-Life in EV to 10–15 years, typically preserving 70–80 % of its initial capacity. Up to 4000 cycles and 10 years for EV use were estimated [12]. In 2017, we assessed 101 cells (100 Ah), finding their State-of-Health (SoH), expressed as % of nominal capacity [13], between 70 % and 95 % after 5.6 and 8.0 years, with an average of 84 % [6]. In 2016, 'Schaufenster Elektromobilität', estimated an End-of-Second-Life (EO2L) of 40–50 % by empirical data, but without any justification [14]. In 2018, Muenzel [14,15] estimated SL use down to 50 % of nominal capacity, whereas more conservative approaches set EO2L at 70 % SoH [16]. Unlike previous practices that relied on static EO2L thresholds, we introduce a dynamic threshold based on residual analysis, as detailed in Section 4.2.4.

We investigated the potential of First-Life EV cells [6] (model:

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Kyburz DXP, eight 100 Ah cells) for SL applications. The generalizability of our findings on lithium iron phosphate (LiFePO₄, LFP) cells to other chemistries is discussed in [Section 1.3.6](#). While the cells' age and mileage from their EV use are known, field data like voltage, current, capacity, and temperature profiles were not recorded. No correlation was found between capacity loss and time in the EV or mileage [6], as analyzed in [Sections 3.1 and 4.1](#).

The same cell type was used in the 'Racing Green Endurance' project to build the longest-range (514 km) road EV in the world 2010, the first EV which has driven the entire length of the 'Pan-American Highway' (26,500 km) [17]. A capacity fade of 5.8 % after 87 cycles was measured and a capacity loss of 32 % after 3000 cycles was estimated. The results are further analyzed in [Section 4.1.1](#).

Recently, Cao et al. [8] studied fresh and decommissioned LFP cells, as well as complete modules. They examined the impact of voltage limits, current, and temperature on degradation over 10'000 cycles within a reduced voltage range (2.8–3.6 V instead of 2.0–3.65 V per datasheet). They support the re-use of the First-Life LFP cells for SL purposes, if the working voltage range remains between 2.80 V and 3.55 V, high working temperature are avoided, and the currents remain in the medium range. Their findings are compared with our cells in [Section 4.1.3](#).

Catton [18] highlighted that stressors like high temperature during calendar aging can induce a History effect, influencing later degradation even under mild conditions. Cells stored at 60 °C, and later moved to 35 °C, experienced long-term degradation ([18], pg. 49 f., 92 f., 151 f.). We broaden their concept of History effect by encompassing all past stressors, including those from storage, transportation and manufacturing. Details are discussed in [Sections 4.1, 4.2 and 4.4](#).

This study aims at modeling the capacity retention over cycle life and comparing the results with existing literature. The report is divided in two main parts:

- (1) Detailed review of aging mechanism and models for LFP battery capacity fade ([Section 1.2](#)), the key background for interpreting our findings in [Section 4](#)
- (2) Experimental data on long time cycling of LFP cells ([Section 3.2](#)), focusing on
 - a. The effect of temperature conditions ([Sections 3.5 and 4.4](#)).
 - b. The relevance of physical modeling for interpreting experimental data ([Section 4.2.4](#))

1.2. Introduction to degradation mechanisms

1.2.1. General

State-of-Charge (SoC) is defined as the current capacity state of a battery cell, either expressed as % of nominal capacity (C_{nom}) or in absolute values as 'Ah'. Depth-of-Discharge (DoD) is defined as the percentage of the cell that has been discharged relative to the nominal capacity. The current rate is commonly expressed as C-rate - the current (A) divided by the rated capacity (Ah).

To understand capacity loss and degradation mechanisms, it is crucial to distinguish between calendar aging (cell in idle state, no current flow) and cycle aging (cell operating under current load). For LFP cells, calendar aging is driven mainly by temperature [19–22], with SoC playing a secondary role below 50 °C [21,23,24]. Cycle aging also depends on charge throughput ([Section 1.3.4](#)) [25]. In EV First-Life use, average current rates are around C/5 (e.g., 20 A for a 100 Ah cell), and EVs spend over 90 % of the time parked. Thus, calendar aging plays a larger role in overall degradation [26].

Strategies to mitigate aging effects and extend battery lifespan are outlined in [Section 1.3.7](#).

1.2.2. Degradation modes during cycling conditions

As SL use involves moderate currents, we focus on capacity fade, and don't consider power fade. The main capacity loss mechanisms— Loss of

Lithium Inventory (LLI; also known as "loss of cyclable lithium" or "primary loss" or "loss of active lithium"), Loss of Active Material (LAM, also known as "secondary loss"), Under-Charge (UC) and Under-Discharge (UD) [27] - often act in coupled, time- and path-dependent ways, complicating cycle life prediction [28]. The role of UC, UD, and polarization resistance is discussed in [Section 1.2.3](#). [Section 1.3.7](#) links these degradation modes to various operating conditions (temperature, current-rates, voltages) across various lithium-ion cell chemistries.

Loss of Lithium Inventory (LLI) refers to the irreversible loss of available lithium ions. The ions are primarily consumed during solid electrolyte interface (SEI) film formation on the anode and continuous thickening mainly as a result of parasitic side reactions, by decomposition of electrolyte [29]. On the one hand, the SEI protects the anode from solvent co-intercalation, preventing exfoliation of the graphene layers; on the other hand, it consumes cyclable lithium. The SEI structure is unstable and undergoes continuous breakdown and reformation, especially under high voltage, or mechanical stress, elevated temperature or varying temperature [27,30]. Each time the SEI layer partially dissolves and reforms, more lithium is consumed in the formation of new SEI compounds [31]. The SEI is generally understood to consist of a compact, highly ion-conductive inner layer and a more porous outer layer that permits solvated lithium ions, but debate remains on whether the electron or the solvent diffusion is the rate determining step, relevant for power capability [32,33]. Early-stage capacity loss in pristine cells is largely attributed to LLI from SEI formation and coupled material decomposition processes [34], with electrode degradation playing a minor role. SEI growth modeling is discussed in [Section 1.3](#).

Loss of Active Material (LAM) refers to the deactivation of available active electrode material necessary for lithium intercalation, caused by various mechanisms [35]. It is important to distinguish between LAM of the cathode (LAM_{cat}) and LAM of the anode (LAM_{an}):

- LAM_{cat}: At high SoC, Ni-rich cathodes (e.g., NMC, NCA) and high-voltage LCO are prone to electrolyte oxidation, primarily triggered by lattice oxygen release, often accompanied by structural degradation and surface exfoliation [36–39]. Formation of a cathode electrolyte interphase (CEI), analogous to the SEI on the anode, passivates the surface but increases impedance and can contribute to LAM. CEI formation is especially significant in Ni-rich and high-voltage cathodes at elevated voltages, but less so in LFP [40–43]. Transition metal dissolution (Mn, Ni, Co, Fe) can lead to metal ion crossover and subsequent deposition on the anode, leading to anode poisoning with irreversible capacity loss [44,45]. Fe dissolution from LFP can occur at >40 °C during extended cycling, or in the presence of HF [46–49].
- LAM_{an}: Solvent Co-Intercalation in graphite can cause exfoliation (layer separation or electrode delamination) [50–52]. Electrolyte decomposition at graphite edges destabilizes the SEI, exposing fresh surfaces and promoting continuous SEI growth during cycling [32,53]. Progressive SEI thickening can clog pores and render parts of the anode electrochemically inactive [21]. Exfoliation also accelerates anode degradation through particle cracking, isolation, and loss of electrical contact [53].
- Mechanical stresses from cycling - such as particle pulverization and binder degradation - contribute to LAM. To increase energy density, graphite anodes often incorporate silicon oxide (<10 wt%); but its large volume changes during cycling (up to +300 % for Si, less for SiO_x, vs. +10 % for graphite) necessitate limiting silicon content to minimize degradation [30,54].

Lithium plating is a critical failure mode, causing rapid capacity loss, and increasing the risk of short-circuits. This phenomenon occurs alongside both LLI and LAM_{an} (e.g., graphite cracking, SEI thickening) [55,56]. Key triggers include anode potential depression at high SoC or low temperatures, combined with kinetic limitations such as fast charging protocols, overcharging conditions, poor N/P ratios, and

electrode geometric misfits [30,57–60]. Even under moderate cycling conditions (e.g., 5–65 % SoC, 23 °C, 0.5C cycle rate), Lithium plating can occur [61]. Calendar aging can also raise the plating risk during subsequent cycling at subzero temperatures (e.g., –5 °C) compared to fresh cells [62].

Copper dissolution from the anode current collector can occur during deep discharge or under rated conditions when moisture or HF is present, significantly reducing cell lifespan. [63,64].

1.2.3. Polarization resistance and cell capacity

Polarization resistances are categorized into resistive (ohmic, “IR loss”) and faradaic (non-ohmic) components. Faradaic polarization includes activation (charge-transfer) and concentration (diffusion) polarizations [65–67]. Resistive polarization typically arises from reduced electronic conductivity due to degradation of cell components, including current collectors, connectors, electrodes and electrolytes [68]. Faradaic polarization, is primarily caused by hindered lithium ion transport due to electrolyte degradation and structural changes in electrodes, such as pore blockage [69].

Electrolyte lithium ion concentration is optimized to support efficient ion transport and maintain performance up to 80 % State-of-Health [70]. Internal resistances can rise due to electrolyte loss caused by solvent evaporation via diffusion through the housing. The role of electrolyte loss through plastic housings is detailed in Section 4.3. This electrolyte depletion accelerates SEI growth, promotes lithium salt precipitation (e.g., LiPF_6), isolates active material particles, and increases electrolyte viscosity, thereby reducing ion mobility. These effects collectively contribute to Loss of Lithium Inventory, Loss of Active Material, and higher resistance, driving capacity fade [70–73].

Polarization resistances can result in Under-Discharge (UD) or Under-Charge (UC) [74]. In UC, incomplete lithium extraction from the positive electrode active material causes the cell to reach the charge cut-off voltage prematurely. In UD, discharge ends before fully de-intercalating lithium from the graphite host. UC can be mitigated by applying a trickle charge (e.g., C/20; floating charge) during the constant voltage (CV) phase, as discussed further in Section 4.2.3 [27].

Extended cycling often leads to LLI, where insufficient lithium is available for full intercalation, resulting in resistance increases. This can cause localized Joule heating, accelerating SEI growth and exacerbating LLI. Calendar aging typically results in a near-linear rise in internal resistance, often linked to ionic resistance increases due to surface layer formation (e.g., SEI growth). This effect is more pronounced at higher temperatures and State-of-Charge values. Cycle aging, on the other hand, can sometimes decrease internal resistance due to increased anode porosity. Both calendar and cycle aging contributed to resistance changes in our cycling experiments [29].

Key methods for assessing internal resistance include Direct Current Resistance (RDC) and Electrochemical Impedance Spectroscopy (EIS). RDC, following IEC 61960 (Section 2.3.2), yields a single resistance value encompassing both ohmic and faradaic components, influenced by current, pulse length and sampling rate [66]. Evaluating potentials at different time scales (Section 3.1.3 in [66]) helps to differentiate these effects. Since both RDC and Incremental Capacity Analysis (ICA, Section 1.2.5) assess potential changes under charge perturbation, similar conclusions about degradation mechanism can be drawn when measured at the same SoC. Correlations between diffusion polarization (concentration overvoltage) and phase transitions over the cell's cycle life have been observed [65]. EIS provide deeper insights by separating resistance into ohmic and faradaic (charge-transfer and diffusion) components. Various models have been developed to link resistance increases to capacity fade [75–77]. When applying discharge pulses of 1–30 s, resistance measurements include contributions from electron transfer (ohmic) and faradaic processes (ion transfer and diffusion) [67,78–80]. Our RDC characterization (Section 2.3.2) employs 10-s discharge pulses, capturing IR loss, activation polarization (charge-transfer), and concentration (diffusion) polarization (detailed in Sections 3.3 and 4.2.4).

1.2.4. Capacity recovery

Capacity loss may be irreversible or reversible, depending on the underlying mechanisms and conditions [20,23,81]. Irreversible loss typically results from side reactions - such as SEI growth or electrolyte decomposition - that primarily affect one electrode, often the anode [23], permanently consuming cyclable lithium or active material. Reversible loss may occur under several conditions: Self-discharge during idle periods involves side reactions at both electrodes [20,23,82]; while initially reversible, it can become increasingly irreversible with prolonged rest [82]. At full SoC, spontaneous re-intercalation of Li ions into the delithiated cathode can cause partial reversible loss. In LFP cathodes - sensitive to lithium variation at high SoC - this may coincide with irreversible electrolyte decomposition [18,20]. Lithium ion diffusion into the anode overhang can temporarily render Li ions inactive, with recovery possible during cycling, sometimes yielding coulombic efficiencies over 100 % [83,84]. Shallow cycling (e.g., 10 % DoD) may lead to recoverable loss when followed by storage at a SoC of 0 % or 100 % [83]. Fast charging (e.g., 4C) can reduce accessible capacity, recoverable by lowering the current to C/10–C/2 [85]. Lithium plating may be reversed through rest at full SoC, allowing re-intercalation of metallic lithium into adjacent graphite [86,87], or through a constant voltage (CV) phase in the CCCV charging protocol, which promotes stripping of plated lithium [88].

1.2.5. Analysis of degradation modes by Incremental Capacity Analysis

The charge curve (V-Q plot) shows cell voltage as a function of time or capacity under constant current, reflecting the cell's chemistry and aging mechanism through distinct voltage plateaus, which correspond to phase transitions in the active electrode materials. Incremental Capacity Analysis (ICA), also known as Differential Capacity Analysis (DCA), enhances interpretation of these phase transitions by plotting the inverse derivative of cell potential with respect to charge capacity

(dQ/dV -V-plot) [89–91], providing insights into mechanisms like LLI, LAM, and Lithium plating.

In LFP/graphite cells, ICA primarily focuses on the graphite anode, as the LFP cathode exhibits a flat potential at ~ 3.45 V versus Li/Li^+ [92–94]. Lithium ion intercalation in graphite occurs in multiple stages, each characterized by distinct voltage plateaus [95–99], which appear as peaks in ICA plots with specific shapes, intensities, and positions [27,68].

Peak analysis is most accurate at low currents (e.g., C/25 or below) to minimize peak broadening and preserve features [27]. However, studies demonstrated that ICA remains effective at higher currents, such as C/3 [74], similar to the conditions used in the present study (Section 2.3.3).

Aging typically reduces peak height, broadens peak shape, and shifts peak positions. Broadening at lower potentials during discharge often reflects SEI destabilization or graphite surface disorder [27,100], while shifts during charging indicate increasing polarization, potentially leading to UD and UC [74]. Polarization increases are often SoC-dependent and therefore asymmetric [100].

To diagnose LAM from ICA, a significant decrease in the height and integral of the first peak must be observed (usually alongside with reduction of second peak). This peak corresponds to the initial phase transition in the graphite anode during Lithium-ion intercalation and serves as a key indicator of active material depletion [101] (see Section 4.2.4). Whether Lithium plating results in a visible ICA peak depends on factors like plating reversibility, C-rate, and SoH [91,102–106].

1.3. Introduction to capacity fade modeling

1.3.1. General

Battery aging is complex, modeled using physical, empirical, or semi-empirical approaches. Physical models (e.g., crack-propagation, porous electrode, single-particle, based on first principles and electrochemical equations [17]) are computationally intensive, require extensive

parameterization, and are impractical for real-world use. Empirical models, using simple functions like polynomials, rely on experimental data and statistical analysis. While efficient, they lack insight into degradation mechanisms and have limited extrapolation. Semi-empirical models combine mathematical expressions with aging mechanisms, balancing simplicity and accuracy for better data interpretation.

1.3.2. Temperature-dependency of capacity loss

Capacity loss in Lithium-ion batteries is temperature-dependent, with elevated temperatures accelerating degradation through reactions like SEI formation, electrolyte decomposition and transition metal dissolution [20]. These reactions make lithium ions unavailable for cycling (LLI), contributing to capacity loss. The Arrhenius equation, commonly used to model this dependency, is expressed as:

$$k(T) = A \cdot e^{-\frac{E_a}{RT}} \quad (1)$$

where k is the reaction rate, A is the Arrhenius factor (depending on the process), E_a is the molecular activation energy for the reaction, R is the universal gas constant, and T is the absolute temperature [29,107–109]. While typically used for small cells, we assess its relevance for BESS SL applications.

1.3.3. Capacity fade models for calendar aging for single battery cells

Capacity fade during calendar aging can be described by Eq. (1), with SoC incorporated into the pre-exponential factor A . However, this model may only apply to pristine cells, as it lacks time-dependent parameters. Long-term aging is time-dependent and often follows a square-root relationship (aging proportional to $\sqrt{t}$), particularly linked to SEI layer growth. Zabala et al. [53] presented an adapted empirical model, where α_1 , β_1 , α_2 and β_2 are fitting parameters (Eq. (2)):

$$Q_{\%loss}(T, SOC, t) = \alpha_1 \cdot e^{\beta_1 \cdot \left(\frac{1}{T}\right)} \cdot \alpha_2 \cdot e^{\beta_2 \cdot SOC} \cdot t^{0.5} \quad (2)$$

Capacity loss during calendar aging can evolve over time, with the exponential parameter $z = 0.5$ changing, reflecting the underlying mechanism, as reported by Gasper et al. [109]. Using the power law function (t^z) and other models to describe LFP cell aging, they linked capacity changes to varying solvent diffusion SEI growth rates. The study identified three stages: (i) non-diffusion-limited SEI growth ($z \approx 1$) at the start, (ii) solvent diffusion-limited SEI growth (z decaying to 0.5) as the SEI thickens, and (iii) a final stage with capacity loss below diffusion-limited rates (z declining to 0.45).

The power law model and alternative models were used to describe the long-term calendar aging (up to ~ 3.6 years) of LFP and other cell chemistries at various temperatures and SoC levels [7,76]. For high cycle numbers, capacity fade may accelerate ($z = 0.45$ [109]) or become linear ($z = 1$ [110]).

1.3.4. Capacity fade models for cyclic aging for single battery cells

In addition to solvent-diffusion-based (physico-chemical) SEI growth models, often based on Tafel-like kinetics [81,111], other mechanistic or semi-empirical models have been developed, such as those considering Lithium plating [112] or Loss of Active Material [113]. Given the complexity of capacity modeling, even for inactive cells (calendar aging), we focus on empirical and semi-empirical models, such as linear, square-root, power law, stretched exponential, and sigmoidal models, and hybrid forms [29,109]. These models, however, have limitations, including non-applicability to variations (e.g. operation outside data-sheet specifications or changes in material composition) or being over-simplified (valid only for limited cases), or over-fitted (requiring excessive parameters and computational power).

According to Heyer's critical review [114], the power law model is widely used for describing cycling aging across various chemistries, including LFP (discussed further in Section 1.3.6).

In 2010, Wang et al. [115] adapted the power law model by replacing the time dependence (t^z) from calendar aging with charge throughput (Q^z) for cycle aging. The charge throughput (in Ah) accounts the charge during charging and discharging. For continuous cycling without rest periods, Q^z is approximately proportional to time, allowing both calendar and cyclic aging to be described using a unified framework. We refer to Wang's model [115] from this point forward, as given in Eq. (3):

$$Q_{\%loss}(T, C_{rate}, Q) = b_1 \cdot e^{\left(\frac{b_2 + b_3 \cdot C_{rate}}{RT}\right)} \cdot Q^z \quad (3)$$

$Q_{\%loss}$ describes the capacity loss of the cell, b_1 is a model parameter for each SoC condition, b_2 , b_3 and z are model parameters, Q is the charge throughput of the cell, C_{rate} is the current, T is the absolute temperature of the cell and R is the universal gas constant.

Since the power law model does not account for faster degradation at advanced cycle life, sigmoidal models have been introduced [109,116]. These models combine a linear trend and a logistic function, describing battery degradation in three main stages:

- Initial Cycles: Rapid linear capacity fades due to a very thin SEI layer [109,117]. Though some studies report an initial capacity increase due to a larger DoD along with enlarged graphite d-space and enhanced lithium ion diffusion [118,119].
- Intermediate Stage: As the SEI layer thickens, square-root degradation is typically more suitable [107,120]. In general, capacity loss decreases as the SEI layer becomes thicker.
- EO(2)L: Accelerated degradation due to self-reinforcing mechanisms, such as LAM detected by ICA [53,107]. Some studies, however, suggest decelerated degradation at this stage [116].

Sigmoidal models capture both EOL and EO2L, addressing the faster degradation seen at EO2L, which the power law model misses. They may be useful for modeling remaining useful life, knee-onsets, and sudden capacity drops at EO2L [121,122]. Recently, cubic functions have also been used to fit capacity trajectories [123].

A new model, similar to the curvature of the sigmoidal model, derived from modified Thevenin circuit, Butler-Volmer kinetics and regression analysis was presented [124]. Recently, data-driven degradation estimation techniques were evaluated [125].

1.3.5. Model transfer for stacked battery cells

Many established physical models predict capacity retention are based on single cells and not entire battery packs, often with narrow temperature ranges or low discharge currents, and may overlook factors like unsteady discharge behavior or environmental stress [17]. We treat stacked cells like single cells with slight variations in cell temperature (Section 4.2.2). For aging, the temperature within the inner cell components, rather than the surface temperature, is critical (Section 3.5 and 4.4). Additional stress factors that may overshadow Arrhenius effects (Sections 4.1, 4.2, and 4.4). Our system is more complex, as our cells have experienced unknown First-Life stress, were cycled and characterized with delays, making it difficult to differentiate between calendar and cycle aging. We use a semi-empirical power law model to interpret capacity retention over cycles (Sections 3.2 and 4.2), expressed as:

$$Cap = a + bx^c \quad (4)$$

where Cap is the capacity, x is the full equivalent cycle (FEC) number, a represents the initial capacity at 0 cycles, b is the 'assumed' cycle-independent factor, and c is the material-dependent parameter that describes the degradation. Based on Sui et al. [76], a can be fitted or set as the initial measured capacity, minimally affecting b and c . Gasper et al. [109] found that temperature (Arrhenius equation) and voltage (Tafel plot) influence capacity degradation rates, but are approximately cycle-independent, allowing inclusion in b (β_1 in [109]). This

assumption is examined in Section 4.2.2. Time- and cycle-dependency are integrated in x^c and are not further differentiated within the scope of our work.

1.3.6. Model transfer to other Lithium-ion battery chemistries

Capacity degradation and cycle life vary with cathode and anode materials. Among lithium-ion chemistries, Nickel-manganese-cobalt oxide (NMC) and lithium-iron phosphate (LiFePO₄, LFP) battery cells are commonly used in EVs and ideal for SL applications due to their long cycle life, good thermal performance, and high specific energy [117]. While lithium manganese oxide (LMO) is less common, it is often blended with NMC to enhance performance. Other chemistries in EV include lithium nickel-cobalt-aluminum oxide (NCA, e.g. Tesla) and lithium manganese oxide (LMO). We compare common lithium-ion battery chemistries, ordered by typical cycle life, and summarize dominating capacity fade mechanisms associated with each. Reported values may vary across sources [30]. Note that this table is not exhaustive and excludes emerging chemistries such as solid-state, sodium-ion, lithium-sulfur, silicon-anode, LMFP, LNMO, and others (Table 1). LFP cells were selected for their safety, thermal and chemical stability, environmental benefits, and cost-effectiveness, making them well-suited for EV and SL BESS applications [126].

As noted in Section 1.3.4, the power law model is not limited to LFP and has been applied across various lithium-ion chemistries [114]. Yang et al. [144] assessed multiple models for six SL use cases of LFP cells; including the above-introduced Wang method and the Swierczynski method – the latter using a square-root dependency – and confirming their suitability for modeling capacity trajectories [145].

Liu et al. [146] demonstrated that the power law model effectively captures capacity fade to EOL (80 % SoH) in LFP cells. Wang et al. [115] validated it over a wide test matrix (−30 °C to +60 °C, 90–10 % DoD, and C/2 to 10C). Wu et al. [147] also applied it at low temperatures (−10 °C, −20 °C).

The model's applicability depends on the dominant degradation mechanisms. A decelerated capacity trajectory is typically linked with SEI-driven aging [30], while linear or accelerating trends may require alternative models. Regardless of model choice, the use of knee-onsets, and our proposed residual-based approach is broadly transferable across chemistries (discussed in Section 4.2.4).

1.3.7. Strategies to mitigate degradation to maximize battery lifespan

Many manufacturing guidelines exist [43]. Various degradation mitigations strategies were published [148]. This section focuses on operator-level. Stress factors were ranked by impact as follows: Charge Current > Temperature > Discharge Current > DoD > Average SoC [149].

1.3.7.1. Current.

- High currents can impair anode intercalation kinetics, promoting Lithium plating [150], and excess heat, accelerating SEI growth. However, fast charging at 2C and +45 °C may improve cycle life compared to +25 °C, likely due to enhanced kinetics of lithium insertion and removal from the host lattice [151]. Rapid ion transport can also cause particle cracking, particularly in graphite anodes and high-Ni cathodes [151]. To extend lifespan, current should be limited under moderate temperatures in line with manufacturer specification.

1.3.7.2. Temperature.

- High temperatures raise SEI electronic conductivity, accelerating electrolyte decomposition, SEI growth, and capacity loss [152]. Temperature-dependent SEI formation models exist [153]. Aging from −20 °C to +25 °C is dominated by Lithium plating, while +25 °C to +70 °C promotes SEI growth and Mn dissolution for NMC cells cycled at rates of 1C ([154], 2014; Section 1.2.2). At −20 °C Lithium plating and at +55 °C Iron dissolution was confirmed for cycling LFP cells ([155], 2020). At +5 °C better capacity retention was observed than at −5 °C or +12 °C when cycling LFP at 1C (CCCV) ([156], 2018). The slightly different tipping point of LFP and NMC were mentioned [157]. In 2024, optimal cycle life was reported for cycling NMC/C at +25 °C [158] and for NMC/silicon-graphite anodes at +30 °C [159]. These thresholds vary by cell type and are also influenced by thermal history (History effect) [158]. For cycling LFP, Fe dissolution—negligible at moderate temperatures—accelerates above +60 °C even during rest, leading to LAM via graphite layer blockage [153,160], and may initiate at +40 °C [161]. SEI decomposition can begin at +35 °C [30,152]. Some chemistries suffer mechanical strain and particle cracking below

Table 1
Comparison of most common lithium-ion battery chemistries.

Battery Chemistry	Cycle Life	Primary Degradation Mechanism	Key Benefits	EV use (Market Share)
LFP (LiFePO ₄)	3'000–5'000+ [127]	SEI growth; minimal Fe dissolution under extreme conditions (high C-rates, temperatures, SoC) [40,46,150].	High safety, long cycle life, low cost, moderate energy density (~90–160 Wh/kg). Slow kinetics reduce Lithium plating risk in cold climates.	Mainstream & Budget (~25 %; growing)
NMC (LiNi _x Mn _y Co _z O ₂) (NMC811 (x = 0.8, y = 0.1, z = 0.1) is more common than: NMC622, NMC532, NMC111)	1'000–3'000	Mn and Ni dissolution, electrolyte oxidation, CEI and SEI growth, gas formation; particle cracking / binder breakdown [128–133].	High energy density (~150–250 Wh/kg), mid-range cost, good cycle life.	Mainstream (>50 %)
NCA (LiNiCoAlO ₂)	1'000–2'500	Particle pulverization, SEI growth, gas formation; Ni and Co dissolution [132,134].	High energy density (~200–320 Wh/kg).	Premium (~20 %)
LMO (LiMn ₂ O ₄) (often blended with NMC)	500–1'000	Mn dissolution, Jahn-Teller distortion, electrolyte oxidation [135–137].	High power output, low cost, high thermal stability [138].	Limited (<5 %)
LCO (LiCoO ₂)	500–1'000	CEI, Co dissolution, structural instability, gas generation, electrolyte oxidation [139,140].	High energy density (up to ~240 Wh/kg).	Rare (~1 %)
LTO (Li ₄ Ti ₅ O ₁₂)	5'000–20'000	LAM mainly on cathode; the LTO anode's near-zero strain (~2 %) reduces SEI growth and particle cracking compared to graphite (~10 %) [141–143].	Extremely long cycle life, high safety, exceptional fast-charging capability, resilient under extreme climate conditions.	Niche / Specialty EVs (~1 %)

–27 °C, while others tolerate storage down to –30 °C, with studies showing reduced degradation at lower temperatures (e.g., –10 °C vs. 0 °C) [19].

- In hot climates, charging should be restricted to cooler periods (e.g., at night). Thermal management systems for EVs are reviewed in [162,163]; further strategies are detailed in Sections 4.4.2 and 4.4.3. In cold climates, self-heating designs (internal heating) can enhance cycle life [164,165]. A CV phase in CCCV protocols can mitigate Lithium plating [62]. Multi-stage charging may also help [166].
- In Summary, cells should be operated within manufacturer-specified temperature ranges, and frequent temperature fluctuations should be avoided for maximum longevity [30,167].

1.3.7.3. State-of-Charge.

- At high SoC or voltage, delithiated cathode surfaces exhibit increased reactivity with the electrolyte, promoting electrolyte oxidation and CEI formation. These side reactions degrade electrode materials and contribute to capacity fade, while CEI formation is minor in low-voltage chemistries like LFP [30,168]. During gas formation [169,170], oxidation byproducts such as CO₂ and formaldehyde may migrate to the anode, destabilizing its SEI and accelerating degradation [36,171]. Additionally, the elevated graphite anode potential at high SoC enhances electrolyte reduction and SEI growth, leading to LLI [29].
- Shallow cycling (e.g., 30–50 % DoD) and operation at intermediate SoC levels generally improve battery lifespan compared to deep cycling (>50 % DoD), as partial cycles impose less structural stress on electrodes and avoid voltage extremes [7,40,44].
- A dedicated BMS is essential to prevent both Under-Discharge (UD) and Over-Charge (OC), as these conditions can cause severe and irreversible cell degradation. UD may lead to copper dissolution and electrode decomposition, and cells that have self-discharged below the specified minimum voltage must not be reused and should be safely disposed of. OC must also be strictly avoided, as it can accelerate degradation—particularly through electrolyte oxidation and, in high-voltage chemistries, aluminum dissolution from the cathode current collector. While OC is less critical for LFP cells, it poses a risk for high-voltage cell chemistries [30,172].
- Some studies report no or minimal cycle life benefit from short rest phases after charging [173,174], while others observe slight capacity recovery (Section 1.2.4).
- To minimize calendar aging, storing cells at low SoC (<10 %) is effective [19,175], but carries a risk of UD if self-discharge is not carefully monitored. Therefore, storing at a moderate SoC is generally considered best practice.

1.3.7.4. Humidity/salinity.

- Water contamination during manufacturing or moisture exposure during storage (e.g., via diffusion) can form HF, which degrades electrode materials [63,64,152]. Therefore, storing cells in high-humidity environments – such as refrigerators – should be avoided. LLI has been identified as a dominant aging mechanism accelerated by humid climates [176].

1.3.7.5. Pressure.

- For certain chemistries, applying appropriate external pressure can mitigate capacity degradation, for instance by enhancing contact between the electrode and current collector [177,178]. This is particularly beneficial for silicon–graphite anodes, where pressure can reduce electrode expansion and related capacity fade [178,179]. A detailed analysis of pressure effects on various degradation mechanisms is provided in [180].

1.3.7.6. Mechanical impact.

- Vibration and shock can degrade electrodes; detailed reviews are provided in [181,182].

When defining EO2L levels, temperature effects should be considered, as they can accelerate degradation during idle (storage) periods. BESS manufacturers should either remove EO2L cells or regularly assess their SoH to ensure safety. Safely decommissioned cells from the BESS must be properly stored appropriate monitoring. Various strategies for handling EO(2)L cells are outlined in [183–187].

In summary, extending battery lifespan involves maintaining stable temperatures, keeping cells at mid-range SoC, limiting DoD with voltage cutoffs, reducing current loads and mechanical stress, and using a robust BMS.

2. Methodology

2.1. Battery cells

This study evaluates cells from privately owned electric vehicles (IDs: “priv.1–6”) and postal service EVs (IDs: “post.1–3”) with varying initial capacities. During First-Life, each EV battery pack comprised eight cells in series, assumed to have experienced similar electrical stress [188].

In order to assess the influence of temperature on the aging during Second-Life, we've selected cells of similar capacity of the same EV battery stack for each capacity group. The selection includes:

- “High initial capacity cells” (low degradation): Cells from “priv.1” (TS: 40, 37, 38)
- “Medium initial capacity cells” (medium degradation): Cells from “priv.5” (TS: 65, 66, 67).
- “Low initial capacity cells” (high degradation): Cells from “post.3” (TS: 4, 5, 2).

The complete capacity distribution for all 72 cells, and the 9 selected cells is presented in Section 3.1.

The cells under study are Thunder Sky TS-LFP100AHA (Shenzhen, Guangdong, P.R.C.), commonly used in EVs and power systems [17,189–197]. Each cell has a nominal capacity of $C_{\text{nom}} = 100$ Ah, dimensions of 218x143x67 mm, and a mass of $m = 3.2 \pm 0.1$ kg [198].

To simulate real-world conditions, we assembled three stacks of three cells each, separated by 3 mm fireproof mineral fiber insulation plates (ISOPLAN 1000, Frenzelit GmbH, thermal conductivity of $\lambda = 0.13 \text{ W m}^{-1} \text{ K}^{-1}$ at 400 °C). The stacks were placed in a climate chamber (Fig. 1 for cell position and for calibrated temperature sensors). This setup enables analysis of how the middle cell's slightly elevated temperature affects capacity fade per cycle relative to the outer cells. The thermocouples were mounted on the cell housing using Kapton tape (thickness: 65 μm) or placed in the gas atmosphere.

We tested at climate chamber (Heraeus Vötsch VT4021/S and HC 7020) set points of 50 °C, 30 °C, and 10 °C, with the 10 °C condition in winter seasons typical for basement environments for most of home Battery Energy Storage Systems. For each temperature group, cells were selected from the same vehicle battery stack, all with residual capacities over 70 Ah from their First-Life in EVs.

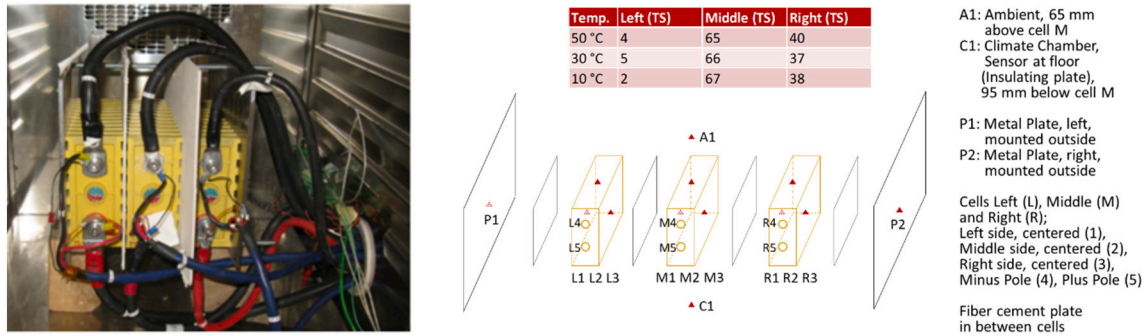


Fig. 1. Left: Test setup of the battery stack. Right: Scheme of the position of the cells (cell ID in table) and the thermocouples.

2.2. Cycling procedure

In this report, cycle count refers to SL discharge-charge cycles in the climate chamber. The cells were electrically cycled, 200 cycles per series (Fig. 2, left) by a constant-current constant-voltage (CCCV) charging protocol with cut-off voltages of $U_{min} = 2.8$ V and $U_{max} = 3.6$ V, and currents as specified by the manufacturer's datasheet ([198]).

Each cycle-series began with a discharge from an initial SoC of 40 Ah at a discharge current of $I_{disch} = 50$ A until the lower discharge cut-off voltage U_{min} was reached, followed by a rest period of $t_{rest} = 5$ min. The cells were then recharged to U_{max} with a current of $I_{ch} = 36$ A and charged at constant voltage until the terminal current of $I_{term} = 5$ A was reached, followed again by a rest period of 5 min.

The duration for which the cells were kept at 20 °C (RDC and capacity procedures) or room temperature (e.g., weight measurements) varied between series, depending on the measurement workload. Calendar aging effects during these periods are acknowledged and discussed in Section 4.1.3.

Initially, current cables were disconnected every 200 cycles for measurements (not included here). Later, disconnections occurred less frequently, during maintenance or weight and temperature assessments. These procedures could influence capacity and RDC values, as discussed in Sections 4.1.2 and 4.2.4.

2.3. Characterization methods

2.3.1. General

Lithium-ion battery performance is typically evaluated by parameters such as capacity, direct current resistance (RDC), power rate capabilities, and energy density [199]. We focused on industrial characterization methods for capacity and RDC measurements, as well as temperature and mass loss.

2.3.2. Discharge capacity and direct current resistance characterization

Every 200 cycles, the climate chamber was set to 20 °C for 5 h to

allow temperature stabilization. Cells were then simultaneously characterized (Fig. 2, middle) according to IEC 61960 (7.2 and 7.7.3) to determine discharge capacity ("Disch. Cap.") and "RDC". The RDC characterization closely followed the widely referenced USABC method (30 s pulse) [66,78,80].

For Discharge capacity measurements, cells at 40 Ah SoC were discharged at 20 A to U_{min} , followed by a rest period of 30 min, then charged at 20 A to U_{max} with a terminal current 5 A, followed by a rest period of 1 h, discharged with 20 A to determine the discharge capacity, followed again by a rest period of 5 min, and then charged with 20 A to a SoC of 20 Ah for the RDC determination.

The RDC procedure involved charging at 20 A to the targeted 40 Ah SoC, resting for 31 s, discharging at $I_1 = 20$ A for 10 s, then at $I_2 = 100$ A for 1 s, resting again for 31 s, and finally charging at 20 A to the next SoC level (20 Ah more, or charged to U_{max}). Cells were finally left at 40 Ah SoC. The RDC value was calculated using:

$$RDC = \frac{U_2 - U_1}{I_2 - I_1} \quad (5)$$

Here, U_1 is the last voltage data point at I_1 , U_2 is the last voltage data point at I_2 . The measurement points 1 and 2 are denoted in Fig. 2, right.

For the first 6200 cycles, including RDC characterization and cycles #6200–6400 (prior to RDC characterization), a four-electrode probe setup was employed, achieving a measurement uncertainty of 1–5 mΩ. For characterizations starting at completed cycle #6400 and subsequent cycling series, an improved custom four-electrode probe setup was implemented. This enhanced setup featured home-built screws to optimize current-free paths and to ensure precise voltage readings near the cell terminals, reducing the expanded measurement uncertainty to below 0.02 mΩ ($k=2$, $\pm 0.01\%$ + 1 digit for the voltage and $\pm 0.02\%$ + 1 digit for the current). Calibration was conducted in compliance with ISO/IEC 17025:2017 standards. RDC-results are presented in Section 3.3, Figs. 8 and 9. Section 4.1.2 discusses potential effects of intermediate RDC characterization every 200 cycles on capacity trends.

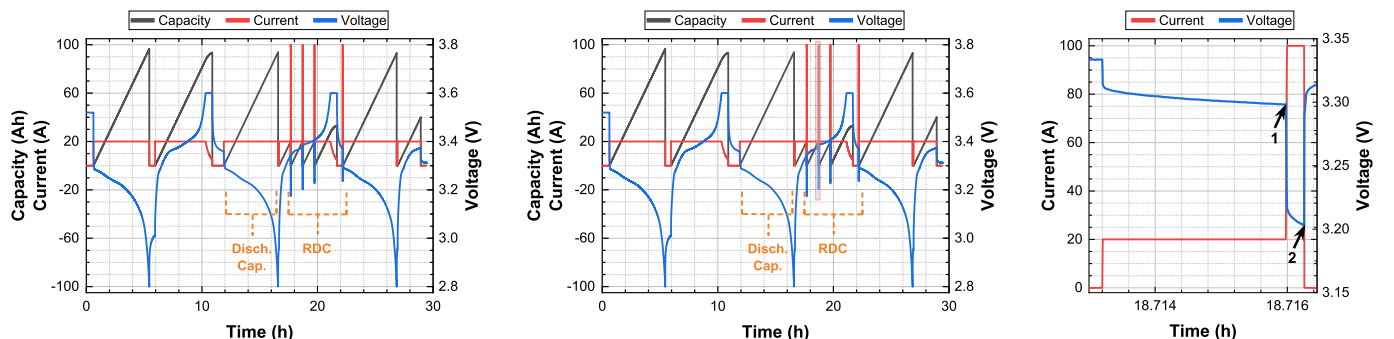


Fig. 2. Left: Cycling procedure. Middle: Procedure to determine the discharge capacity and RDC. Right: Zoom In of Middle.

2.3.3. Incremental Capacity Analysis

Unlike the typical currents of approximately C/25 used for Incremental Capacity Analysis (ICA), we analyzed the voltage data during the charge phase of the cycle series as described in Section 2.2. Although this approach may slightly reduce peak resolution and data accuracy, it allows us to identify and discuss unique degradation features, particularly for TS5, analyzed in Section 4.2.4.

2.3.4. Temperature measurement

Type K thermocouples were used to monitor local temperatures at multiple cell positions. For the 10 °C cell stack, initial measurements were recorded using the auxiliary temperature channels of the Maccor S4000 system, as detailed in Section 3.5.1. For independent measurements a custom-built multiplexed temperature recording system was used, comprising a Keithley 2700 multimeter and two NTCs reference sensors, stabilized in an aluminum block to ensure accurate readings (see Section 3.5.2).

Thermography recordings (Section 3.5.3) were conducted using the Infratec ImageIR 8380 HP S camera system. During these recordings, an additional TS-LFP100AHA cell was used. Type K thermocouples were attached to the backside of this cell while leaving the front side exposed for thermal imaging. To achieve more accurate spatial temperature measurements, the climate chamber ventilation was disabled during the thermography recordings, minimizing airflow-related temperature variations.

2.3.5. Mass measurement

Cell masses were recorded using a balance (KERN KB 3600-2 N; results in Section 3.6).

3. Results

3.1. Results from First-Life

Fig. 3 shows First-Life data on the relationship between mileage SoH (directly corresponds to capacity in Ah). The variation in SoH among cells within the same EV block, which experienced identical charge-throughputs and current loads, is of particular interest. The findings are discussed in Section 4.1.1.

3.2. Capacity

3.2.1. Charge capacity during cycling

Table 2 presents the charge capacity during the cycling series, analyzed in Section 4.1.4. As each series begins with a discharge step from a SoC of 40 Ah, rather than a fully charged state, only the capacity

data for the charge phase is reported. Capacity values for cells cycled at 10 °C, 30 °C and 50 °C differ from those at 20 °C, as detailed in Section 4.1.2. Furthermore, calendar aging between cycle #0 (initial characterization in March 2017) and cycle #1 (cycling starting in August 2017) must be considered.

3.2.2. Discharge capacity during RDC/capacity-characterization

Despite the Arrhenius equation's relevance in modeling aging, no significant temperature-related distinctions in the discharge capacity recorded at 20 °C were observed initially (analyzed further in Section 4.4). Cycling continued with the 10 °C and 30 °C stacks, focusing later solely on the 30 °C stack.

We analyzed capacity trajectories with respect to both Cycles (Fig. 4) and Full Equivalent Cycles (FECs, Fig. 5), with the reasoning detailed in Section 4.1.2. Residual analysis confirmed the power law model (Eq. (4)) as a suitable fit without additional terms. Parameters a , b , and c were estimated using a nonlinear Gauss-Newton algorithm with trust region methods ('lsqcurvefit' in Matlab), starting with initial guesses of 80, 1, 1. For Fig. 5, the charge throughput per 200-cycle series was calculated as the average capacity of the 1st and 200th cycle multiplied by 200.

Data for TS66 at cycles #5600, #5800, #6000, #6200, and all cells at cycle #8800 were discarded due to measurement errors. Key time points (Fig. 4 top right; discussed in Sections 4.1.2, 4.2.1 and 4.2.2) include:

- (1) Maintenance delay (88 days) after characterization at cycle #0, visible at #200.
- (2) Maintenance delay (126 days) after cycle #2000, visible at #2200.
- (3) Maintenance delay (85 days) after cycle #4800, visible at #5000.
- (4) Customized four-electrode probe setup implemented after cycle #6400, visible at #6600.

Capacity trajectories were renormalized based on SoH. Instead of normalizing to 100 % initial capacity, the trajectories were horizontally shifted, resetting cycle counts to zero at 70 Ah, just below the lowest initial capacity after First-Life (new axis: 'Renormalized Cycles' respectively 'Renormalized FECs'). Data with negative cycle numbers were discarded. Eq. (4) was then refitted with a constrained to 70 Ah, and parameters b' and c' were re-determined, summarized in Tables 3 and 4.

The results from Figs. 4 and 5 are summarized in Tables 3 and 4. Capacity trends are discussed in Sections 4.2.2 and 4.2.4, along with the red-marked TS5 RMSD value.

For the statistical analysis, we focus on Table 4.

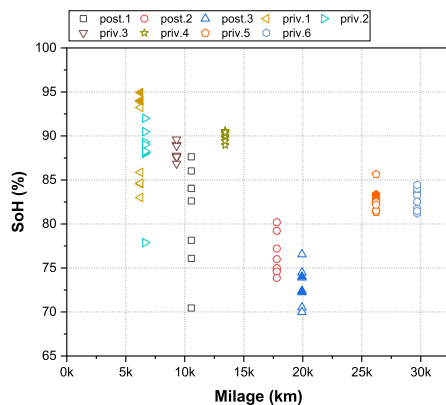


Fig. 3. Left: Capacity retention after First-Life vs. mileage for nine EVs, each with battery stacks of eight cells each (multiple overlapping data points). SoH of all 72 cells was measured in March 2017, with cells selected for the SL experiment marked by filled symbols. Right: Age data of the selected cells. The table lists the vehicle ID ('EV no.'), mileage (km), cell production date, and the start and end dates of EV use. 'EV (yr)' indicates years in EV, and 'age (yr)' is the cell's age as of March 2017.

Table 2

Charge capacities for cells ('ID TS') disassembled 'from EV no.' (Fig. 3). Capacities at cycles #0 and #200 are from the RDC-procedure (Fig. 2, middle) at 20 °C. Charge capacities of 1st and 50th cycles, recorded during cycling at the set-point temperature ('temp. class'), and their absolute capacity loss in Ah are summarized in last column.

EV no.	ID TS	in. cap. group	Cap. #0 (Ah, @20°C)	Cap. #200 (Ah, @20°C)	temp. class	Cap. 1st cycle (Ah)	Cap. 50th cycle (Ah)	Cap. loss 1 st -50 th (Ah or %Cnom)
priv.1	40	high	94.0	82.1	50 °C	95.2	91.4	3.8
priv.5	65	medium	83.2	74.9	50 °C	83.2	80.5	2.7
post.3	4	low	72.4	65.8	50 °C	75.5	72.9	2.6
priv.1	37	high	93.9	85.0	30 °C	92.8	90.0	2.8
priv.5	66	medium	82.7	76.5	30 °C	80.9	79.2	1.7
post.3	5	low	74.0	68.3	30 °C	72.5	71.0	1.5
priv.1	38	high	94.9	85.2	10 °C	83.0	79.6	3.4
priv.5	67	medium	83.4	75.7	10 °C	73.6	71.1	2.5
post.3	2	low	72.2	63.8	10 °C	61.3	57.7	3.6

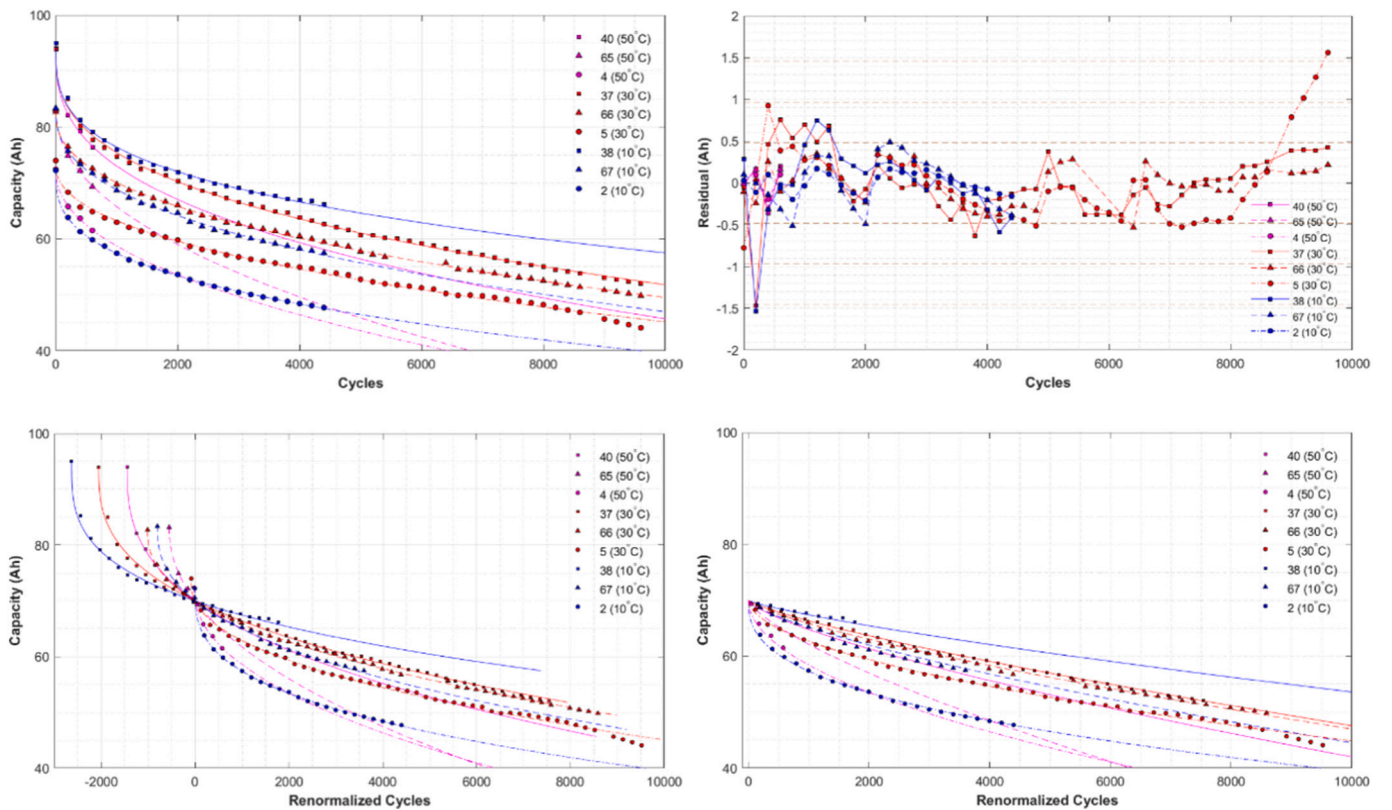


Fig. 4. Capacity versus cycle number for the three cell stacks at different temperatures (50 °C: magenta; 30 °C: red; 10 °C: blue) and initial capacities (high: solid, medium: dashed, low: dash-dotted). *Top left:* Raw capacity data. *Top right:* Residuals with ± 1 , ± 2 , ± 3 SD for TS5; positive residuals indicate faster degradation than predicted. *Bottom left:* Capacity after coordinate transformation. *Bottom right:* Fitting vectors after truncating the cycle number vector, showing minor deviations from bottom left. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

First, we test whether lower initial capacity (column 'Par. a (Ah)') correlates with higher degradation (columns 'dCap(Ah)'). A negative Spearman's rank correlation coefficient (ρ) confirms this inverse relationship with statistical significance ($p < 0.05$) at all three SoH levels of 70, 60, and 50 Ah.

Second, we examine whether higher temperature (column 'temp. class') is associated with increased degradation. Linear fits across all three SoH levels (Fig. 6) reveal a statistically significant negative correlation ($p < 0.05$) only in the high initial capacity group at 70 Ah.

A key objective of a future Battery Management System (BMS) with integrated State-of-Health estimation is the early detection of deviations

from the power law model. Therefore, additional fits were performed using only 4 or 10 data points and extrapolated (Fig. 7; results discussed in Section 4.2.3). The confidence intervals were calculated using a non-linear Levenberg-Marquardt algorithm (Software: Origin). For more degrees of freedom for the fit and improved confidence interval predictions, parameter a was constrained to the initial capacity at 0 cycles.

3.3. Direct Current Resistance

The RDC values at a SoC of 40 Ah are shown in Fig. 8.

The non-linear increase in RDC for TS5, beginning at cycle #7200,

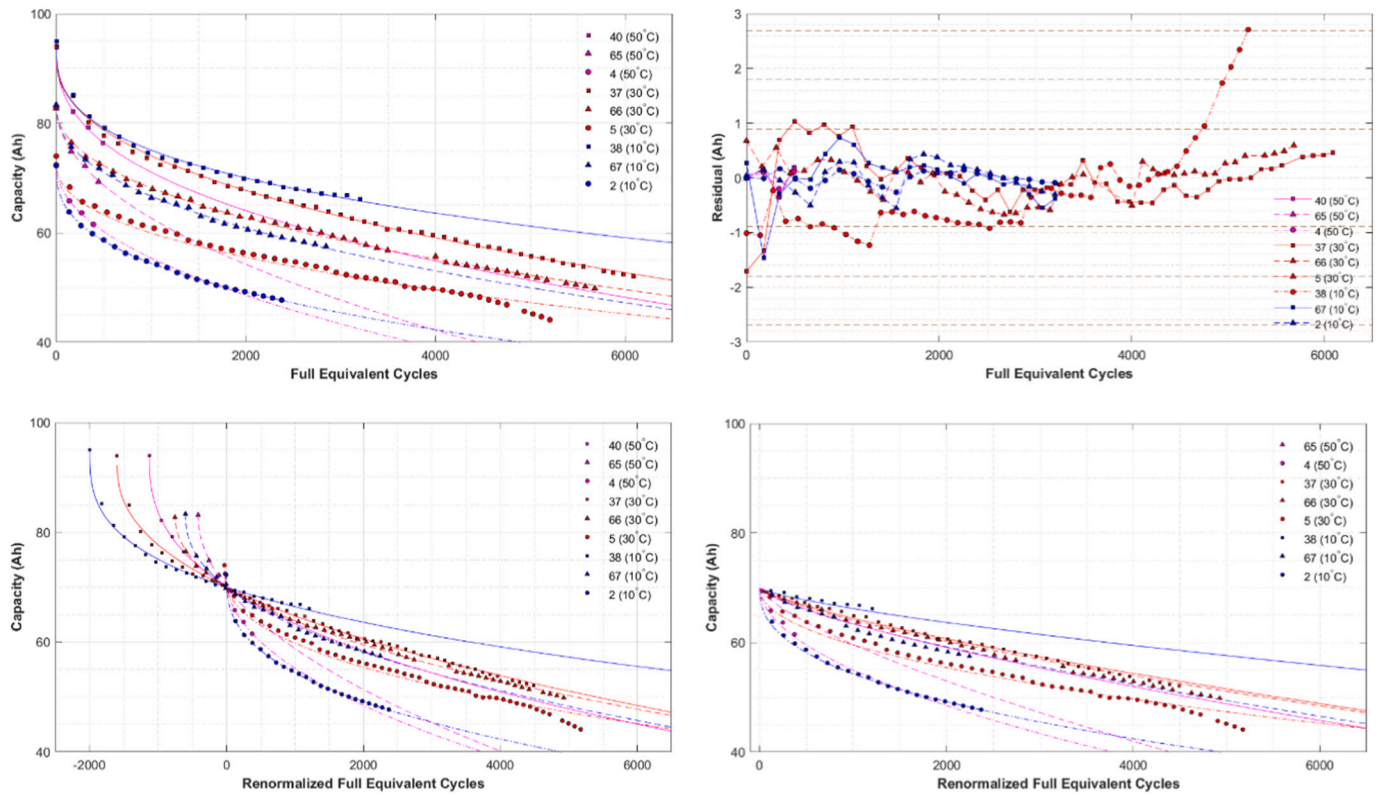


Fig. 5. Capacity trajectories versus Full Equivalent Cycles and corresponding residuals. Descriptions follow those in Fig. 4.

Table 3

Fitting parameters and root mean squared deviation (RMSD) for the capacity data. The position in the relative stack of the Cell ID is given in Fig. 1. Cycle loss (dCap) in Ah per 1000 cycles is based on the calculated parameters a , b , and c , from Fig. 4.

ID	TS	temp.	in. cap.	class	group	Par. from experimental data					Par. from fit points		
						a	b	c	n-3	RMSD	dCap	dCap	dCap
						(Ah)	(Ah)	(-)		(Ah)	70 Ah	60 Ah	50 Ah
40	50 °C	high				94.0	-1.72	0.36	1	0.44	6	3	2
65	50 °C	medium				83.1	-0.66	0.47	1	0.38	11	6	4
4	50 °C	low				72.4	-0.56	0.46	1	0.24	48	7	4
37	30 °C	high				93.9	-1.56	0.36	45	0.40	4	2	1
66	30 °C	medium				82.6	-0.68	0.42	41	0.24	5	2	1
5	30 °C	low				73.2	-0.45	0.45	45	0.50	18	3	2
38	10 °C	high				95.2	-2.35	0.30	20	0.48	3	1	1
67	10 °C	medium				83.4	-0.95	0.40	20	0.32	7	3	2
2	10 °C	low				72.3	-1.39	0.34	20	0.13	186	7	2

and more pronounced from cycle #8000 (4466 FECs) on, is discussed in Section 4.2.4.

To compare RDC data across different SoC levels, we define $RDC = R_0 + dR$ for better visualization of changes. The difference dR is plotted against the cycle number in Fig. 9. Between cycles #6400 and #6600, we observed a sudden increase of 0.15 mΩ in one of the three cells, which remained notably high in subsequent cycles. Thus, dR is set to zero at 6600 cycles. For cycles #6600–9600, all three cells exhibited capacities between 40 and 60 Ah, allowing for three RDC measurements per cell: at 20 Ah, 40 Ah, and fully charged (as outlined in Section 2.3.2). The results will be further analyzed in Section 4.2.4.

3.4. Incremental capacity analysis of TS5

Due to observed ambiguities in the capacity plot, RMSD-plot, and RDC-plot, we include an Incremental Capacity Analysis (ICA) plot in Fig. 10 for further discussion in Section 4.2.4.

3.5. Temperature

3.5.1. 30 °C cell stack

At cycles #4800 to #5000, the 30 °C group cells were acclimated for over 5 h before cycling began. We evaluated the temperature one hour before the cycling start, with all calibrated thermocouples measuring 30.2 ± 0.1 °C (137 data points per hour per channel, 2 SD). Fig. 11 displays the temperature data.

Table 4

Capacity data with respect to Full Equivalent Cycles (FECs), from Fig. 5, obtained by the same procedure as for Table 3.

					Par. from experimental data							Par. from fit points		
ID	temp.	in. cap.			<i>a</i>	<i>b</i>	<i>c</i>	n-3	RMSD	dCap	dCap	dCap	<i>b</i> '	<i>c</i> '
TS	class	group			(Ah)	(Ah)	(-)		(Ah)	70 Ah	60 Ah	50 Ah	(Ah)	(-)
40	50 °C	high			94.0	-1.59	0.39	1	0.45	8	5	3	-0.043	0.73
65	50 °C	medium			83.1	-0.64	0.50	1	0.40	16	9	6	-0.073	0.72
4	50 °C	low			72.4	-0.59	0.49	1	0.25	65	12	6	-0.406	0.52
37	30 °C	high			92.2	-0.90	0.43	45	0.53	6	4	3	-0.032	0.75
66	30 °C	medium			83.4	-0.70	0.45	41	0.38	8	4	3	-0.040	0.72
5	30 °C	low			73.0	-0.70	0.42	45	0.95	41	6	3	-0.372	0.48
38	10 °C	high			95.2	-2.12	0.33	20	0.46	4	2	1	-0.024	0.73
67	10 °C	medium			83.4	-0.85	0.43	20	0.30	10	5	3	-0.050	0.71
2	10 °C	low			72.2	-1.34	0.37	20	0.12	212	12	5	-0.949	0.41

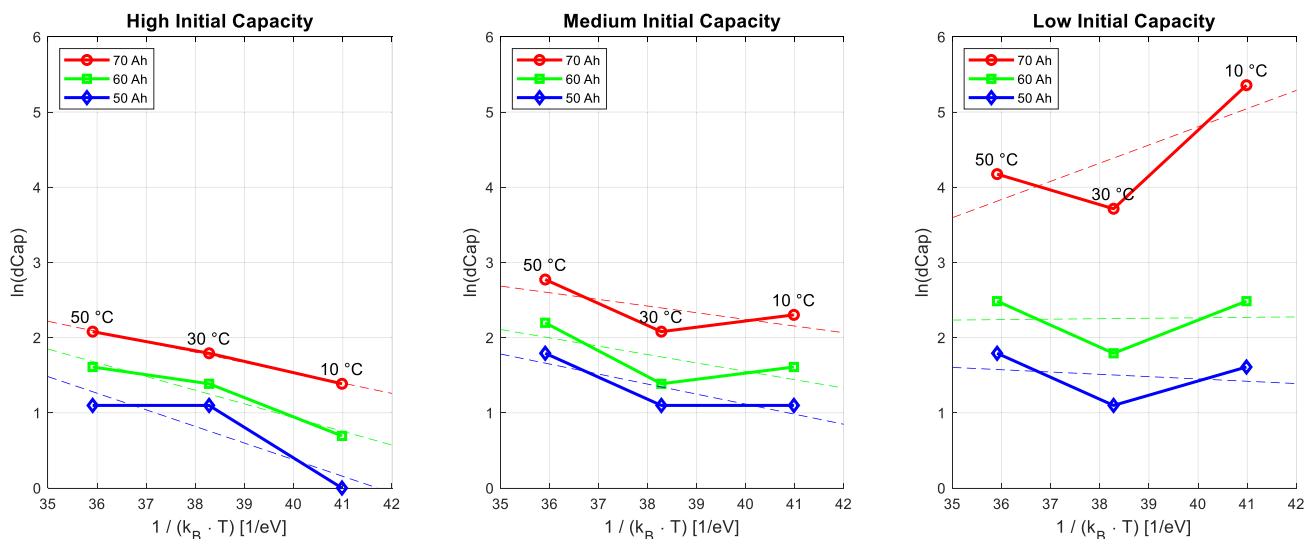


Fig. 6. Arrhenius Plot for the three different initial capacity cell groups. The dashed lines represent the linear fits.

The measured cell temperatures varied between 32.5 and 34.5 °C at L2, M2, and R2. Temperatures between cells ranged from 35.0 to 38.0 °C for L3/M1 and M3/R1. The left cell completed its 200 cycles first (15.07.2021, 20:28), and a reduction in temperature oscillations was observed at 20:51. Temperature coupling between adjacent cells introduced complexities in predicting cell aging. The impact of neighboring cell activity on the cell's capacity and temperature is discussed in Section 4.2.2.

3.5.2. 10 °C cell stack

Before the cycling series #200–400, the 10 °C cells were acclimated for 3 h at 10.0 °C. One hour before cycling, sensors at positions L1, M1, and R1 (Fig. 1) measured ambient (8.4 ± 0.1 °C), TS2 (9.1 ± 0.1 °C), TS67 (9.1 ± 0.2 °C), and TS38 (9.0 ± 0.2 °C). Discharge for all three cells began simultaneously. Temperature and capacity trends over time for the 10 °C stack are illustrated in Fig. 12, with further analysis provided in Section 4.4.2.

During the second discharge cycle, the left and right cells' capacity increased by 1 Ah, while the middle cell increased by 2 Ah compared to the first cycle. The capacity change seems coupled to temperature: the left cell (TS2) completed 200 cycles with a maximal temperature of 14.5 °C (sensor L1).

The end of the cycles, indicated by the first cell's completion (left cell, TS2), is apparent in the capacity reduction of the middle cell, alongside a

temperature drop from 21 °C to 17 °C (sensor M1).

Once the middle cell (TS67) completed its cycles, the right cell (TS38) exhibited a capacity drop, accompanied by a temperature decrease from 19 °C to 16 °C (sensor R1).

3.5.3. Local distribution of heat

To assess thermal diffusion and heat distribution, spatial temperature measurements were recorded during discharges to $U_{\min} = 2.95$ V. Prior to each experiment, an 88.4 Ah cell (not part of the main nine cells), was fully charged to $U_{\max} = 3.6$ V, with a terminal current of $I_{\text{term}} = 5$ A, and left to cool overnight to ambient temperature. No external cooling was applied during the experiments.

In the first experiment, the fully charged cell was discharged at a constant current of 100 A, until the lower cut-off voltage was reached after 44 min, with temperatures rising from 25.7 to 25.9 °C initially to 32–38 °C by the end (Fig. 13, middle).

In the second experiment, the fully charged cell was discharged at 300 W constant power (93 A initial current). Initial IR camera readings showed 26.9 ± 0.2 °C ($k = 2$) across the cell. A spatial temperature gradient up to 2 °C was observed within 20 min (Fig. 13, right).

3.6. Mass loss

Fig. 14 presents the mass loss over time for the 10 °C and 30 °C

Grad Series	50°C			10°C			30°C		
TS	□ 40	□ 65	□ 4	□ 38	□ 67	□ 2	□ 37	□ 66	□ 5
const	94.0262	83.1589	72.3987	94.9396	83.3504	72.2355	93.9163	82.7234	74.0121
b	-1.5955	-0.6461	-0.5883	-1.4628	-0.8364	-1.3717	-1.3144	-0.6878	-0.8697
$u(b)$	0.2925	0.1420	0.0991	0.1000	0.0767	0.0484	0.2175	0.0847	0.0782
c	0.3859	0.5002	0.4869	0.3797	0.4325	0.3705	0.3972	0.4552	0.3920
$u(c)$	0.0310	0.0376	0.0295	0.0100	0.0135	0.0054	0.0242	0.0182	0.0135
D.F. (n - 2)	2			8			8		

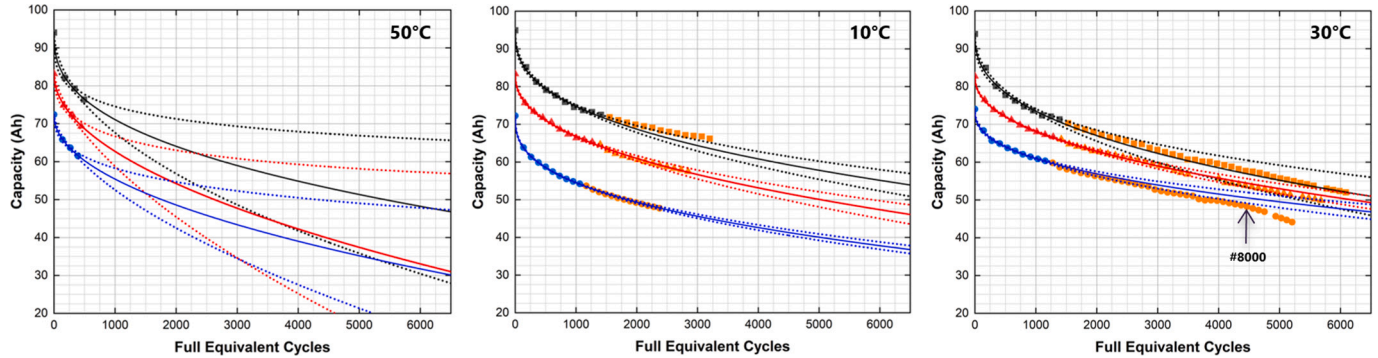


Fig. 7. Capacity trajectories versus Full Equivalent Cycles. The fit lines (solid) and the 95 % confidence bounds (dotted) are displayed. Orange data points are masked and not used for the fits. Left: 50 °C cells, fits from the 4 first data points. Middle: 10 °C cells, fits from the 10 first data points. Right: 30 °C cells, fits from the 10 first data points. Annotation: TS5 at #8000 (4466 FECs).

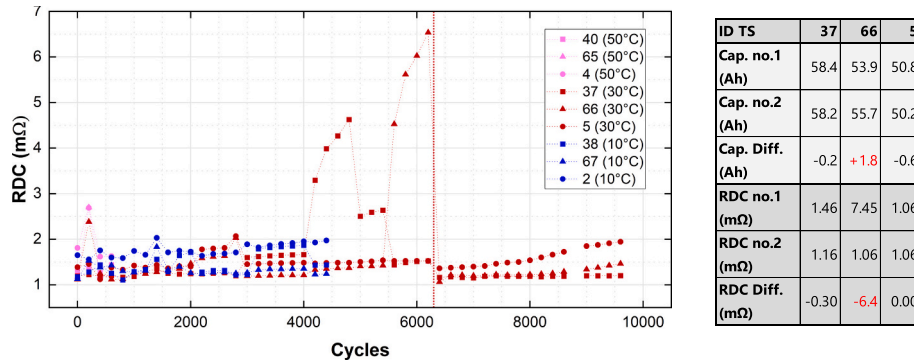


Fig. 8. Left: RDC versus cycle number at 40 Ah SoC and 20 °C. The vertical line at 6300 marks the transition to the new measurement setup. Right: Comparison of Capacity ('Cap.', Ah) and RDC (mΩ) for the original (no. 1) and improved (no. 2) measurement setups for the 30 °C stack. Red numbers indicate high differences for TS66. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

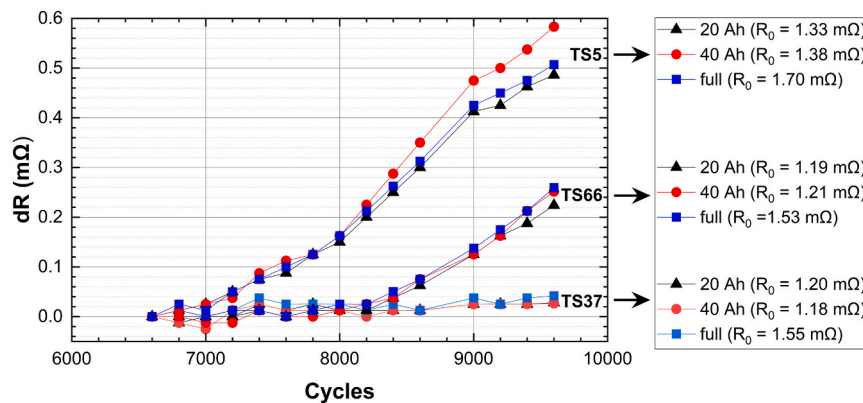


Fig. 9. Difference dR vs cycles, determined at three different SoC levels: 20 Ah, 40 Ah and full. R_0 is determined at 6600 cycles.

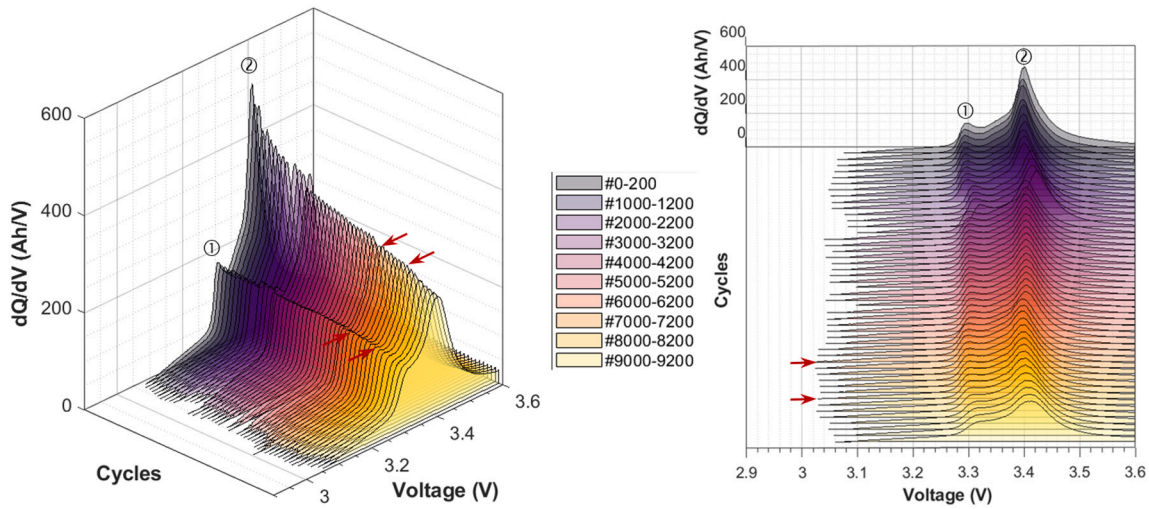


Fig. 10. ICA Plot of TS5. All 48 ICA trajectories (Cycles #0-200 to #9400-9600) are shown with a reduced legend. Red arrows indicate cycles #6800-7000 and #8000-8200. The numbers identify the location of the first peak (①) and the second peak (②). Left: Visualization highlighting the decrease in peak amplitudes over cycles. Right: Visualization of the shifts in the position of the peaks, providing insights into the progression of ohmic and faradaic effects. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

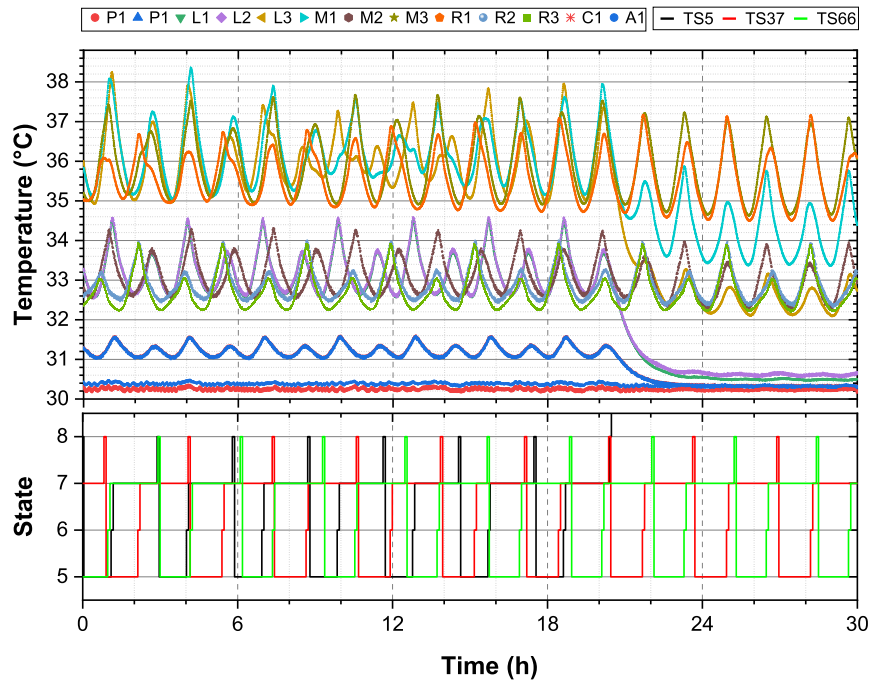


Fig. 11. Temperature values during 30 h of cycling for the 30 °C stack (e.g. TS37: #4973-#4982). Only the left and middle cells are shown for clarity. States: 5 = Discharge, 6 = Rest, 7 = Charge, 8 = Rest, >8 (10/11) = Completed. For better visualization, moving average (data points $i-3$ to $i+3$) were applied to display the ambient temperature for sensors C1 and A1.

stacks, further analyzed in [Section 4.3](#).

4. Discussion

4.1. Capacity degradation

4.1.1. First-Life cycles estimation

A reliable baseline capacity is essential. While the datasheet specifies a nominal capacity of $100 \text{ Ah} \pm 5\%$ [198], reported values range between 107 and 113 Ah [17]. Our 2013 baseline tests (unpublished), conducted at 20 °C using a CCCV protocol ($U_{\min} = 2.8 \text{ V}$, $U_{\max} = 3.6 \text{ V}$, $I_{\text{ch}} = 25 \text{ A}$, $I_{\text{term}} = 1 \text{ A}$) aligned with these specifications. For consistency,

we conservatively set $C_{\text{nom}} = 100 \text{ Ah}$ as initial capacity.

Three methods were evaluated to estimate the FECs experienced by the cells during First-Life:

1. FECs using Mileage data of First-Life:

Assuming 22 km/day over 250 delivery days/year [200] and an energy use of 8 kWh/100 km [201], a postal EV consumes $\sim 440 \text{ kWh}$ annually. With eight 100 Ah cells per vehicle, this corresponds to ~ 162 FECs per cell annually. The “mileage estimated FECs” (Table 5) were calculated using this value and a conversion factor of $1 \text{ Wh} \approx 3.4 \text{ Ah}$. Due to weak correlation between mileage and capacity fade (Fig. 3), alternative FEC estimation methods were explored:

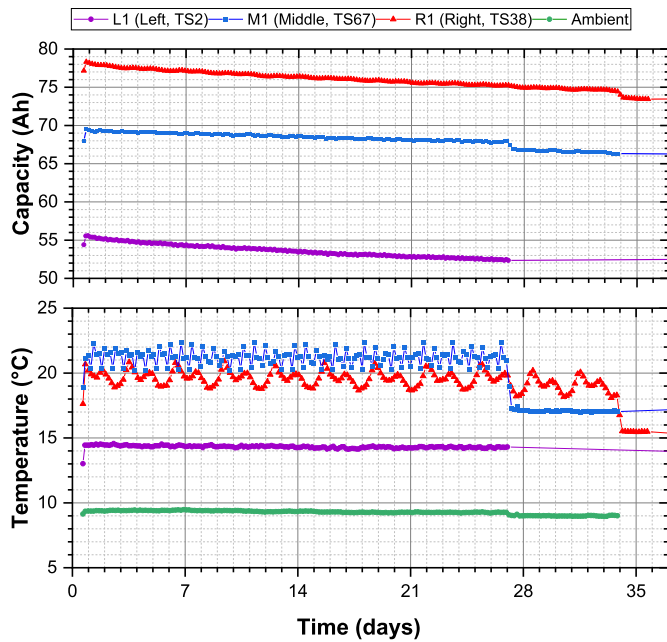


Fig. 12. Capacity and temperature data for the 10 °C stack during cycles #200-#400. Each data point represents the maximum value of each cycle, depicted as a single point. Temperature sensors were positioned at L1 (magenta circle), M1 (blue square), R1 (red diamond), and the ambient temperature in the climate chamber is shown as the mean value of sensors X and Y. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2. FECs using Power Law Extrapolation of SL Data:

To align with an initial capacity of 100 Ah, an artificial value at 0 FECs and an offset (FEC_0) were introduced. The power law model was fitted for FEC_0 = from 0 to 1000. The optimal FEC_0 for each cell

was derived by minimizing RMSD for 1001 fits per cell, (Table 5, “best fit FEC”). However, results were unrealistic for low-milage cells (e.g., 1 FEC for 6’233 km in EV “priv.1” and 19’936 km in EV “post.3”), suggesting early aging deviates from lab-derived power law behavior.

3. FECs using Linear Extrapolation of SL Data:

Using the first two SL data points (0 and 200 cycles), FECs during First-Life were estimated via linear extrapolation (Table 5, “linearization FECs”).

Although the cells were in use for only 4.9 years in EVs, calendar aging over the 7.5 years since manufacturing – when the cells were mostly idle – likely explains the small difference between “milage

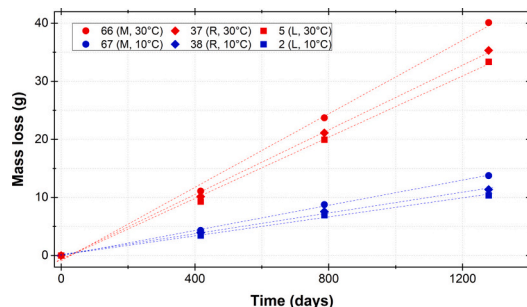
Table 5

Estimation of FECs for First-Life (from First-Life and Second-Life data). Cells identified with Cell ID from: EV priv.1 (6’233 km), EV priv.5 (26’212 km), EV post.3 (19’936 km). The last columns represent a linearized extrapolated value of cycles from First-Life, assuming all cells have measured 100 Ah at the beginning.

EV nr.	ID	temp. class	in. cap. group	Cap. in. (Ah)	milage estimat. FECs	best fit FECs	linearization FECs
priv.1	40	50 °C	high	94.0	183	3	88
priv.5	65	50 °C	medium	83.2	771	31	321
post.3	4	50 °C	low	72.4	586	35	579
priv.1	37	30 °C	high	93.9	183	1	122
priv.5	66	30 °C	medium	82.7	771	46	441
post.3	5	30 °C	low	74.0	586	1	653
priv.1	38	10 °C	high	94.9	183	2	93
priv.5	67	10 °C	medium	83.4	771	54	346
post.3	2	10 °C	low	72.2	586	36	449



Fig. 13. Left: Experimental setup (thermocouples at the back). Middle: First experiment: Thermocouple data after having reached the final voltage of 2.95 V (44 min with 100 A). Right: Second experiment: IR image after 20 min discharge (300 W).



ID	temp. cell		m	q	r
TS	class	pos.			
66	30 °C	middle (M)	0.0317	-0.9	0.998
37	30 °C	right (R)	0.0278	-0.6	0.999
5	30 °C	left (L)	0.0264	-0.7	0.999
67	10 °C	middle (M)	0.0109	0.0	1.000
38	10 °C	right (R)	0.0089	0.2	0.998
2	10 °C	left (L)	0.0082	0.1	0.998

Fig. 14. Mass loss over time. Parameters of linear fit $dm(x) = mx + q$ with dm : loss (g), x : time (days), r : Pearson's r are provided.

estimat. FECs" and "linearization FECs" for EV "post.3" (586 vs. 560 FECs). For "priv.1" and "priv.5", milage estimated FECs exceeded linearization estimates by at least 50 %, likely due to milder operating conditions in private use compared to postal EVs, which involve more start-stop dynamics [202].

Unlike EVs, which often face idle periods and calendar aging, aging in our experiments was predominantly cyclic with minimal calendar effects (Section 4.1.2).

We conclude that linear extrapolation is more suitable than power law fits for limited cycles. However, power law fits remain valuable for extended SL cycling and effectively detect abnormalities, such as potential knee-points, in capacity trajectories.

We benchmarked our observed degradation against Lorf et al. [17], who drove the Pan-American Highway (26'500 km in 90 days) first (in 2010) using identical TS-LFP100AHA cells. They reported a 5.8 % capacity drop over 87 partial cycles (characterized at 21.5 °C, 1C) and predicted 32 % capacity loss after 3000 cycles using rainfall counting ([17], Fig. 9), equating to 0.22 Ah loss per 1000 km.

In our study (Fig. 3), capacity loss per 1000 km was:

- priv.1 (6233 km, 7.3 years): 1.72 ± 1.66 Ah ($k = 2$)
- post.3 (19,936 km, 7.5 years): 1.35 ± 0.22 Ah ($k = 2$)
- priv.5 (26,212 km, 7.7 years): 0.66 ± 0.10 Ah ($k = 2$)

Postal EVs showed slightly higher capacity losses than private EVs (Fig. 3). The higher loss and variation in "priv.1" may reflect differences in cell quality during cell manufacturing. Compared to Lorf et al., our observed degradation was higher, likely due to differing duty cycles (e. g., more start-stop driving) and significantly more calendar aging (85 months vs. 3 months).

4.1.2. The role of calendar aging

Comparing capacity data across studies requires consistent conditions - particularly current, temperature, and voltage limits. As Table 2 shows, lower capacities at 10 °C reflect higher RDC than at 20 °C. Cells cycled at 30 °C displayed reduced capacity in their first cycle compared to 0-cycle data at 20 °C, likely due to calendar aging during the 5-month storage before cycling.

The power law describes calendar aging during storage with a t^2 relation and cyclic aging with a Q^2 relation, both reflecting underlying molecular degradation processes (Sections 1.3.3 and 1.3.4).

Initial measurements showed that cells TS5 and TS2 lost ~3.2 Ah over 5 months of storage at ~20 °C. If extrapolated linearly, this would imply a 59 % loss over 7.5 years. However, observed long-term losses were smaller, consistent with the known deceleration of calendar aging. Interestingly, cells stored at 50 °C aged more slowly (~0.5 % /month), indicating the impact of cell quality and SoH.

Capacity trajectories over time revealed that maintenance delays of up to 126 days resulted in minimal capacity losses (0–1 Ah), well within measurement residuals. By comparison, high-capacity cells lost approximately 5 Ah during 400 cycles over 100 days. Performing one RDC characterization (~30 h) every ~200 cycles (~50 days initially for high-capacity cells) showed negligible impact, confirming that calendar aging has a minor effect on long-term trends compared to cyclic aging.

4.1.3. Capacity fade dependency on State-of-Charge

We investigate whether prior stressors, like cycling at high voltage, cause a History effect that continues to impact capacity loss after the stressor is removed. Cao et al. [8] analyzed aging using comparable CALB CA100 cells (3.2 V 100 Ah LiFePO₄) under varying cut-off voltages and currents:

- 1.) Simultaneous changes to lower cut-off (3.1 V to 2.8 V) and upper cut-off (3.45 V to 3.55 V) had negligible impact (Cycle ~#3000-#4000, Cell 01), indicating aging predominantly occurs above

3.55 V. Cell 03 exhibited accelerated fade when the lower cut-off was reduced to 2.6 V after cycle ~#7000.

- 2.) Lowering the cut-off voltage to 2.6 V (Cycles #5100-#5700, Cell 01) significantly increased the aging rate by +50 % (Cycles #5100-#7100). Interestingly, the fade rate did not slow after resetting the lower cut-off to 2.8 V at cycle #6000. Capacity partially recovered after cycle #7100.
- 3.) Raising the C-rate during charging from 0.6 to 0.9 (Cycles ~#7600-#9900, Cell 01) resulted in a rapid increase in the aging rate by approximately +200 %.

From these findings, we conclude:

- 1.) Cut-off Limits: Accelerated aging was observed below 2.8 V and above 3.55 V. To maximize FECs for Second-Life purposes, maintaining an upper voltage limit of 3.6 V is recommended.
- 2.) SEI layer Stabilization: Capacity recovery after cycles ~#7100 (Cell 01) suggests that SEI layer formation may stabilize aging over time. Even under sustained stressors, capacity loss rates can decrease, and partial capacity recovery may occur. However, the findings from Cao et al. do not provide sufficient evidence for a persistent History effect.
- 3.) C-rate: Changes in the C-rate significantly influence degradation rates, corroborating established knowledge.

Fig. 4 explores the hypothesis that capacity loss is proportional to cycle count, based on the idea that degradation is accelerated when the SoC frequently crosses high levels (cycle-dependency). To limit degradation, conservative cut-off voltages of $U_{min} = 2.8$ V and $U_{max} = 3.6$ V were used, ensuring nearly 100 % DoD while avoiding the more extreme values of 2.5 V or 4.0 V noted in the datasheet. Based on this and the study by Cao et al., we assume that the capacity loss in the upper and lower voltage segments is not more severe than during average SoC segments. This is consistent with the observation that LFP cells are less sensitive to higher SoC compared to high-voltage chemistries (Section 1.3.7).

As indicated in Section 4.1.2, calendar aging plays a minor role, and the effect of capacity loss at our maximal upper voltage limits is not pronounced. We hypothesize that capacity loss is primarily proportional to charge throughput, as evaluated in terms of Full Equivalent Cycles (FECs), illustrated in Fig. 5. This focus guides the subsequent analysis.

Both cycle-based (Fig. 4) and charge-throughput-based (Fig. 5) evaluations are valid approaches. The choice of method depends on whether capacity loss occurs uniformly throughout the cycle and the impact of resting periods between cycles.

4.1.4. Comparison of the capacity losses per cycle with other laboratory studies

Tredeau and Salameh [203] reported a 2 Ah capacity loss (1.25 % of nominal capacity) after the first 50 cycles for TS-LFP160AHA cells ($I_{ch} = C/2$, $I_{disch} = C/2$, $U_{min} = 2.8$ V, $U_{max} = 4.2$ V). In comparison, our TS-LFP100AHA cells ($I_{ch} = 0.36C$, $I_{disch} = C/2$, $U_{min} = 2.8$ V, $U_{max} = 3.6$ V) exhibited a higher capacity loss of 1.5 % to 2.6 % within the same period (50 cycles), despite using lower C-rates and similar voltage cut-offs (Table 2). We approach their reported upper voltage limit of $U_{max} = 4.2$ V with caution, as the TS-LFP160AHA datasheet [204] states risk of cell damage if operated outside 2.5 V to 4.0 V, a range reasonable for Yttrium-doped LFP (LiFeYPO₄). The capacity loss discrepancy may stem from pre-existing stressors from First-Life, such as mechanical and thermal conditions or varied current loads. These findings emphasize how First-Life usage and environmental factors (e.g. shocks, vibrations, temperature fluctuations, humidity, and current loads) influence capacity retention and aging rates.

We also compared our findings with Cao et al. [8], who cycled LFP cells and modules about 10'000 times. Their fresh cell (Cell 01) lost about 33 % of its initial capacity, like our 42 % loss for TS37. However,

Cao's cells operated for under four years at 23 °C, while TS37 had over 14 years (with temperatures up to 30 °C), which likely contributed to the higher capacity loss in our study.

4.2. Power law model

4.2.1. Fit parameters and suitability of the power law model

At 70 Ah, cells with lower initial capacity exhibited higher capacity loss per cycle within each temperature group (10 °C, 30 °C, 50 °C), but no consistent temperature trend was observed across stacks (Fig. 5, bottom right). The power exponent c' was ~ 0.8 for medium/high-initial-capacity cells, and 0.4–0.5 for low-initial-capacity cells (Table 4). TS2's fade rate fell below the diffusion-limited threshold of 0.5, suggesting additional aging mechanisms beyond SEI-related LLI [109].

Fig. 15 illustrates the relationship between RMSD and initial capacity (parameter a , Table 4), revealing that higher initial capacities correlated with larger RMSD, except for cell TS5. This suggests that lower initial capacities may produce better fits due to reduced influence from external factors.

Additionally, the decline in c' and reduced residuals in intermediate stages suggest that early-stage degradation dynamics may change over time.

Evidence for a History effect influencing cell aging rates at different cell life stages includes:

1. Capacity Spread after First-Life (Fig. 3):

Despite identical charge throughput among eight cells per EV during First-Life, significant capacity variations were observed. This likely reflects differences in manufacturing quality, which influenced capacity fade during First-Life usage.

2. Accelerated Decay at the 70 Ah Benchmark:

Cells with faster First-Life degradation exhibited higher capacity fade rate at the 70 Ah benchmark, highlighting the enduring influence of First-Life stress on performance also during SL.

4.2.2. Methodological model limitations

Different models (linear, square-root, hybrid) were tested (data not included here) but yielded no improvements: The power law model provided accurate fits without additional terms. Fit parameter a turned out to be unnecessary and was replaced by the initial capacity in Fig. 7, reducing parameters to b and c .

As noted in Section 1.3.5, temperature and voltage, both identified as time- and cycle-independent, primarily influenced b , with minimal impact on c :

- Temperature Effects: Table 4 shows that b' reflected temperature effects, while c' showed minimal correlation, especially for high- and medium-capacity cells.

- Voltage Effects: Higher cell potentials, associated with higher calendar aging rates ([109], Section 1.3.1), affected b more than c . Long-term cycling introduces variability due to changes in cell potential with SoC, but had negligible effects on both parameters.

In the following we analyze the residual plots (Figs. 4 and 5):

- Cycles #8000–#9600 : TS5: Linked to accelerated capacity loss, initiated at cycle #8000 (4466 FECs; Section 4.2.4).
- The declining residual (~ -1.5 Ah) at cycle #200, the sharp rises (up to $\sim +1.0$ Ah) at cycle #2200 and #5000, and the corresponding trends of falling residuals, likely reflects capacity loss during calendar aging in idle states. The earlier four-probe setup had limitations, which were resolved with an improved setup from cycle #6400 onward (only for characterization at 20 °C). No capacity deviations were observed during cycles #4200–#6200 at 20 °C, likely due to the trickle charge applied during the constant voltage phase (Section 1.2.3) and the use of a 20 A discharge current (Section 2.3.2).
- The earlier four-probe setup had limitations, which were resolved with an improved setup from cycle #6400 onward (only for characterization at 20 °C). No capacity deviations were observed during cycles #4200–#6200 at 20 °C, likely due to the trickle charge applied during the constant voltage phase (Section 1.2.3) and the use of a 20 A discharge current (Section 2.3.2).

Example data from cell TS66 illustrates the charge profile prior entering the CV phase at 3.6 V:

- Cycle #5900 (Charge capacity: 49.70 Ah): thereof 10.64 Ah during CV phase over 33 min
- Cycle #6100 (Charge capacity: 47.58 Ah): thereof 11.46 Ah during CV phase over 34 min
- Cycle #6300 (Charge capacity: 43.86 Ah): thereof 13.32 Ah during CV phase over 35 min
- Cycle #6500 (Charge capacity: 55.43 Ah): thereof 5.64 Ah during CV phase over 23 min
- Cycle #6700 (Charge capacity: 54.93 Ah): thereof 5.55 Ah during CV phase over 22 min

At cycle #6500, the cell reached 3.402 V after charging 30.54 Ah at 36 A - similar to the capacity at the start of the CV phase (3.6 V) during cycle #6300. This corresponds to a ~ 5.5 m Ω resistance difference, consistent with the RDC drop between cycles #6200 and #6400 (Fig. 8).

The CV phase's trickle charge only partially compensates for external resistances, which can cause UC and UD. These higher resistances reduce DoD by reaching cut-off voltages limits earlier and potentially slowing degradation. To address related issues, the improved four-electrode probe setup resolved earlier measurement ambiguities, such as:

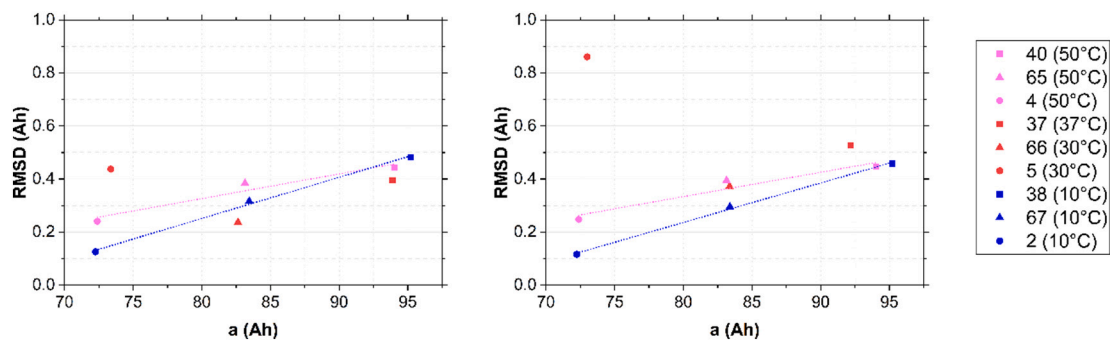


Fig. 15. RMSD vs. fit parameter a (close to initial capacity) for all nine cells (The dotted lines are the linearization (omitted for the 30 °C cells due to the very high values of the two cells with lowest initial capacity). The RMSD value of TS5 (red cursive in Tables 3 and 4) is discussed in Section 4. Left: Data from Table 3 or Fig. 4, Right: Data from Table 4 or Fig. 5. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

- Terminal/cable oxidation or contamination.
- Loose connections (e.g., screws).
- Temperature deviations during characterization (e.g., heat transfer from neighbour cells).

The capacity data (Section 3.2.2, Figs. 4 and 5) validate the power law model (Section 1.3) across battery levels (cells, stacks, and modules), provided that temperature variations are considered (Sections 3.5 and 4.4).

4.2.3. Implications for prognostic EO2L models

A key indicator of increasing degradation rates is the deviation of residuals from the power law model fit, as illustrated in Figs. 4 and 5. Although the model demonstrates strong overall accuracy, exceptions like TS38 and TS5 emphasize the need for constant monitoring. TS38 aged more slowly than predicted, while TS5 showed accelerated capacity loss beginning at cycle #8000 (4466 FECs), clearly visible in Fig. 6 (right panel). These findings underscore the value of integrating prognostic models into Battery Management Systems, which can enable early detection of anomalies.

To determine how many data points are needed for reliable long-term predictions, we analyzed the 50 °C series using just four data points (up to 600 cycles). This led to wide confidence intervals, showing that early predictions can be unreliable. Using 10 data points provides more robust results, especially for the 10 °C and 30 °C datasets. High initial capacity cells such as TS37 and TS38 exhibited unstable early decay trends, making predictions based on only four data points unreliable. In contrast, medium- and low-capacity cells degraded more consistently, and even four data points gave good fits up to 9600 cycles, with only slight improvements when more data were added.

Combining the power law model with complementary techniques like ICA and RDC can improve prediction accuracy. Early detection of anomalous aging supports safer and more efficient BMS. While TS38 aged slightly slower than anticipated, TS5 showed accelerated capacity loss starting from cycle #3000, a trend particularly noticeable in Fig. 7 (right panel).

4.2.4. Insights from TS5

The capacity trajectories for TS5, plotted against both Cycles (Fig. 4) and Full Equivalent Cycles (Fig. 5), reveal notable anomalies. Specifically, the capacity residuals of TS5 (Fig. 5, top right) display a nearly linear increase at cycle #8000 (4466 FECs, 48 Ah) and elevated RMSD values that deviate from the trend observed in Table 4. This indicates limitations of the power law model for long-term predictions. A sigmoidal or cubic model (Section 1.3.4) may better capture TS5's aging trajectory, though additional data is necessary to substantiate this. Nevertheless, power law model residual and RMSD analyses remain effective for detecting anomalies such as self-accelerating capacity fade.

Capacity trajectory predictions based on 10 data points (Fig. 7, right) suggest that the onset of self-accelerating capacity fade could be detected as early as cycle #8000 (4466 FECs). The cycling series for all cells was concluded once the capacity residual for TS5 exceeded the ± 3 standard deviation (SD) threshold (Figs. 4 and 5 top-right), which we propose as a potential early indicator of End-of-Second-Life (EO2L). This threshold is based on the three-sigma rule, a commonly used statistical method for outlier detection in process control. We suggest that this residual-based approach may also be applicable to other cell types, regardless of chemistry or degradation model, even if they do not follow the empirical power-law model. This methodology provides an alternative to EO2L definitions based on knee-onset points [121,122] or static capacity-based thresholds (e.g., 50 %). However, further data and validation are required to assess the robustness and generalizability of this early warning criterion.

Reliable RDC data, collected from cycle #6400 onward, reveals a non-linear increase in RDC values starting at cycle #7200 (Fig. 8), suggesting accelerated degradation mechanism.

Analysis of RDC values across different State-of-Charge levels provides further insights. RDC at 40 Ah SoC shows a significant increase compared to 20 Ah and fully charged, particularly over the last 1000 cycles (Fig. 9). Uniform RDC increases, as observed for TS37, suggest pure ohmic polarization (e.g., electrode tab corrosion). In contrast, TS5 shows a pronounced spread among the three dR lines starting at cycle #8000, indicative of increasing faradaic polarization, likely driven by degradation of internal cell components (e.g., electrolyte, active materials, binders, additives). The primary RDC shifts are attributed to ohmic effects, with faradaic polarization contributing increasingly post-cycle #8000.

Incremental Capacity Analysis provides further evidence of TS5's capacity degradation mechanisms. The ICA plot of the charging process (Fig. 10) highlights: Peaks shifts are primarily caused by ohmic effects, as indicated by a nearly uniform shift to the right ($U_{\text{shift,ohmic}} = \text{RDC} (U_{\text{pos}}) \cdot 36 \text{ A}$). Elevated resistance in earlier cycles (pre-#6400) is attributed to additional ohmic contributions from cables and measurement artifacts. Accelerating polarization shifts are observed from cycle #6800 onward. Minimal relative polarization changes between the two primary peak maxima $\sim 3.3 \text{ V}$ and $\sim 3.4 \text{ V}$ suggest limited faradaic processes. Peak amplitude decrease:

- Second peak ($\sim 3.4 \text{ V}$): A deceleration in peak decline during the first 2600 cycles suggest stabilization of the SEI layer (Section 1.2). Post-cycle #2800, the peak's intensity declines significantly, while the first peak ($\sim 3.3 \text{ V}$) shows a more gradual decrease, reflecting ongoing capacity loss primarily due to LLI. This nearly linear intensity decline aligns with power law degradation patterns observed in advanced lifespans.
- First peak ($\sim 3.3 \text{ V}$): Shows minimal intensity decrease up to cycle #6800, indicating negligible LAM. After cycle #6800, the peak amplitude declines rapidly, accelerating further from cycle #8000 onward, reflecting increasing LAM contributions to capacity loss.
- No additional peaks: The absence of new peaks in the ICA plot suggests that Lithium plating was not a significant contributor to capacity loss in the final cycles.

The observed Capacity trajectories (Figs. 4, 5, and 7) and the RDC-data (Figs. 8 and 9) corroborate ICA findings (Fig. 10). A consistent decline in the first peak's amplitude, beginning around cycle #6800, accelerating post-cycle #8000, indicates LAM and increasing faradaic polarization.

These findings of accelerated capacity loss align with established lithium-ion degradation mechanisms (Section 1.2), emphasizing the importance for close monitoring of TS5 as its SoH approaches 50 %. Establishing defining End-of-Second-Life thresholds is critical to mitigate risks associated with Lithium plating, LAM and ensure operational safety.

4.3. Mass loss

Gas diffusion through the plastic housing follows Fick's law, and the extent of mass loss (Fig. 14, Section 3.6) depends on several factors:

- 1.) Temperature: Higher temperature resulted in higher mass losses ($30^\circ\text{C} > 10^\circ\text{C}$).
- 2.) Cell Position: Middle cells, experiencing heat accumulation due to their location, displayed the highest mass loss within the same temperature class.
- 3.) Capacity Dependence: Higher-capacity cells, which generate more Joule heating (e.g., the right cell compared to the left in Fig. 14), exhibited greater mass loss.

Electrolyte depletion (Section 1.2.3) may accelerate capacity decline via mechanisms like LLI and LAM. However, post-mortem analysis is necessary to confirm whether the observed LAM in TS5's later cycles

(Section 4.2.4) is linked to electrolyte loss. Additionally, electrolyte diffusion-driven material decomposition may further exacerbate mass loss rates.

Future studies must assess the long-term effects of electrolyte loss due to degradation and diffusion, especially for SL applications. To address mass loss, the manufacturer has transitioned to prismatic cells with metal cans, replacing the 4 mm thick polypropylene casing.

4.4. Temperature

4.4.1. Influence of temperature on capacity fade

Capacity fade is primarily attributed by three factors:

1. History effect: Cells that degrade rapidly early on tend to continue degrading faster throughout their lifespan.
2. Arrhenius trend: Higher temperatures accelerate degradation and capacity fade.
3. Other degradation mechanisms: Factors like LAM or Lithium plating may compete with or amplify the History effect and Arrhenius trend.

Analysis of capacity fade at SoH levels of 70, 60, and 50 Ah (Table 4) showed that degradation rates increase with lower initial capacity: high < medium < low. This trend is most evident at 70 Ah and remains statistically significant at 60 and 50 Ah ($p < 0.05$). This supports the History effect hypothesis: Cells with slower degrading near Beginning-of-Life (~100 Ah) experience lower capacity losses with age.

For high-capacity cells, degradation at 70 Ah follows a linear Arrhenius trend (Fig. 6), with degradation accelerating at higher temperature: $10\text{ }^{\circ}\text{C} < 30\text{ }^{\circ}\text{C} < 50\text{ }^{\circ}\text{C}$. However, at lower SoH levels, a reversion between $10\text{ }^{\circ}\text{C}$ and $30\text{ }^{\circ}\text{C}$ emerges, indicating an additional degradation mechanism at later stages. This reversion is more pronounced in medium and low-capacity cells, appearing as early as 70 Ah - possibly triggered during First-Life or manufacturing. High-capacity cells appear more resilient to this effect. The reversion aligns with findings by Waldmann et al. [154], suggesting increased degradation at low temperatures due to mechanisms like LAM or Lithium plating (Section 1.2.2). While no evidence for Lithium plating was found in the ICA analysis (Sections 3.4 and 4.2.4), further post-mortem analysis is needed to confirm the occurrence of Lithium plating or LAM.

The highest fade rate was observed in the combination of low temperature ($10\text{ }^{\circ}\text{C}$) and low initial capacity (0.212 Ah/cycle at 70 Ah for cell 'TS2'), suggesting a synergistic degradation effect from First-Life and temperature stress.

Overall, initial cell condition appears to have a stronger influence on aging than temperature alone. In some cases, the History effect may compete with the expected Arrhenius trend.

Minor temperature differences between middle and outer cells were observed, no significant capacity loss difference was detected at the same cycle number. However, such thermal variations may become more relevant in future aging studies - particularly under extreme SoC conditions or in larger battery packs with limited thermal insulation between neighboring cells.

4.4.2. Effect of neighboring activity on the cell's capacity and temperature

Neighboring cell activity affects the temperature of adjacent cells, with external temperatures ranging from $14\text{ }^{\circ}\text{C}$ to $21\text{ }^{\circ}\text{C}$. These variations complicate the use of simple Arrhenius-based models due to uneven heat distribution.

Middle cells showed a slight increase in discharge capacity - up to 1 Ah - when neighboring cells were active. This is likely due to a small temperature rise, reducing internal resistance, and allowing deeper DoD. However, the impact of this temperature-driven effect was minor compared to dominant aging factors such as First-Life history (History effect).

The effect of neighboring cell activity varies with stacking and cooling strategies. In our tests, cells were electrically isolated, and no

significant temperature-driven degradation patterns were observed. However, in air-cooled, parallel-connected cells, studies have reported faster degradation of middle cells due to higher temperatures and C-rates [205]. Conversely, liquid-cooled systems can result in faster degradation in outer cells, driven by smaller temperature increases and larger thermal gradients, leading to Lithium plating and reduced porosity [206].

These findings highlight the importance of thermal management in larger battery packs to prevent uneven aging and ensure uniform cell performance.

4.4.3. Spatial temperature distributions

Thermal imaging and thermocouple measurements (Section 3.5.3) revealed spatial temperature gradients within individual cells. These gradients become more significant as cells age, particularly in cells with reduced capacity and lower charge transport efficiency [207]. Such non-uniform temperature profiles may accelerate localized degradation, becoming increasingly critical with larger electrode formats. As high-energy-density designs emerge (e.g., BYD's blade battery), effective heat dissipation and advanced thermal management systems are essential [205–208].

To better understand the impact of temperature variation on State-of-Health, further post-mortem analysis is recommended. Coin cells assembled from different electrode regions could reveal spatial degradation patterns, such as SEI growth variations, and support improved cell design and thermal management in large-format battery packs.

5. Conclusions

The tested traction battery cells demonstrated excellent durability under demanding conditions. At $30\text{ }^{\circ}\text{C}$, the cells completed 9600 cycles (~6000 FECs) while retaining over 40 Ah capacity after 14 years of calendar aging, including 11 years of active operation, with an average capacity loss of 35 Ah. While aging slowed during early Second-Life (SL) use, it accelerated for cell TS5 below 50 Ah SoH.

The empirical power law model reliably predicted capacity degradation. Long-term prognosis was feasible using only 10 data points (up to 1800 cycles), enabling early detection of anomalies. Cell TS5, for example, began deviating from predicted behavior around cycle #8000 (4466 FECs), marking the onset of rapid degradation. Capacity residuals exceeding 3 SD threshold may serve as a potential EO2L criterion, emphasizing the importance of continuous monitoring in SL applications to ensure safety.

A statistically significant correlation was observed between lower initial capacity and higher degradation rates. This History effect was most evident at SoH levels of 70 Ah and 60 Ah, with degradation consistently following the pattern: low < medium < high initial capacity, across all test temperatures.

At 70 Ah SoH, high-capacity cells exhibited a linear Arrhenius-type degradation trend ($10\text{ }^{\circ}\text{C} < 30\text{ }^{\circ}\text{C} < 50\text{ }^{\circ}\text{C}$). In contrast, medium- and low-capacity cells showed a reversed trend between $10\text{ }^{\circ}\text{C}$ and $30\text{ }^{\circ}\text{C}$, suggesting accelerated degradation at lower temperatures—potentially driven by mechanisms such as LAM or Lithium plating, which may become more pronounced at low SoH levels. Synergistically, the combination of low initial capacity and $10\text{ }^{\circ}\text{C}$ resulted in the highest fade rate at 70 % SoH.

Heat coupling raised temperatures to $22\text{ }^{\circ}\text{C}$ in the $10\text{ }^{\circ}\text{C}$ stack and $38\text{ }^{\circ}\text{C}$ in the $30\text{ }^{\circ}\text{C}$ stack, emphasizing the need of uniform temperature control. Localized thermal effects may be clarified through post-mortem analysis and coin cells assembled from electrode samples at different positions.

Evaluation of varying resting periods at different SoH levels showed that calendar aging diminished over time and had limited impact on long-term capacity fade compared to cyclic aging.

RDC analysis highlighted the need for current-free voltage measurements. Customized four-electrode setups ensured accuracy for our

low-impedance cells.

Degradation in TS5 was confirmed by RDC, capacity trends, and ICA across SoC levels. ICA identified LAM even at $\sim C/3$ rates, demonstrating its potential with routine cycling data. Further techniques like EIS could elucidate underlying faradaic processes.

Mass loss was in accordance with cell temperatures. Addressing electrolyte loss is critical, particularly regarding Lithium plating risks and Lithium salt precipitation. Post-mortem studies can assess LAM, Lithium plating, and residual electrolyte volume.

To enhance SL safety, we propose integrating a residual-based cut-off threshold for EO2L into BMS strategies. Close monitoring of internal resistance and temperature data near EO2L will further ensure functional safety and performance.

CRediT authorship contribution statement

Daniel Aeppli: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Erwin Hack:** Writing – review & editing, Formal analysis. **Marcel Held:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors report financial support was provided by SFOE. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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